THE BLACK BOOK

Part Test-1 Top-10 Reports

Complete reference: every number, chart, rule and source explained

If you want to know what anything in the impact reports or the institute-vs-our-top-10 comparisons means, how it was calculated, or where it came from, it is in this book. Every explanation is worked through on the real Class 8 reports — **Pragya G8 (EXI) Achievers**, JEE Mains and JEE Advanced — and every number quoted here is computed from the same data as the reports.

Campus
HYD/Nallakunta DD Colony

Tests
JEE Mains 08 Oct 2026 · JEE Advanced 09 Oct 2026

Covers
12 reports · Classes 8, 9 and 10

Prepared
07 Nov 2026

How to use this book

The book is in seven parts. Parts B and C walk through the two kinds of report page by page: each screenshot from the real Class 8 report has ● numbered markers, and the list under it explains each number — **what it shows**, **how it is worked out**, and the **Class 8 value**. Part D collects every formula and threshold in one place, Part E is an A–Z glossary, Part F explains the tool that builds the reports, and Part G lists known data issues and answers to common questions.

> **If you only have two minutes**
>
> - **Page 1 of an impact report** tells you how many marks the top 10 lost, in which subject, and the headline problems.
> - **Section 4** is the to-do list: every question lost by 5 or more of the top 10, worst first, by subject.
> - **Section 7** (the question cards) is what to put in front of a subject teacher: the question itself plus who got it wrong or skipped it.
> - **The comparison report** answers "how far behind the institute's best are our best, and on which questions?".

Contents

A The basics

A1 What you have · A2 Who is analysed · A3 Right, wrong, skipped and "lost" · A4 Paper patterns and marking · A5 Where the data comes from · A6 How a report is made · A7 How the final key is applied

B The impact report, section by section

B1 Header, headline and tiles · B2 Where the damage happened · B3 Takeaways and the two diagnosis boxes · B4 Question-by-question chart · B5 Chapter hot-spots · B6 High-impact questions at a glance · B7 Who lost which question · B8 Subject scores · B9 The notes block · B10 Question cards · B11 What is different in Advanced

C The comparison report

C1 What is compared · C2 Headline and tiles · C3 Subject by subject · C4 Question-by-question gap · C5 Chapter gaps · C6 Biggest gaps list · C7 Question cards · C8 Special cases (overlap, identical groups)

D Formulas and thresholds

D1 Every formula · D2 Every threshold and where it lives in the tool

E Glossary A–Z

Every term, label and symbol used in the reports

F Behind the scenes: the tool

F1 Folder map · F2 What each file does · F3 Paper settings · F4 How question images are cut · F5 Automatic checks · F6 Rebuilding and the next Part Test

G Data notes and FAQ

G1 Known limitations · G2 Questions people have asked

PART A

The basics

What the reports are, who is in them, what the paper patterns are, where every piece of data comes from, and how the final answer key changes things.

A1 What you have

For every class and every exam there are **two PDFs**, so 3 classes × 2 exams × 2 = **12 reports**. They live in `FINAL REPORTS/G8`, `G9` and `G10` (the build also writes them to `MOST IMPACT QUESTIONS/G8…G10`).

- **Top-10 impact report** (`…_Top10_Impact_Report.pdf`) – where our campus's top 10 students lost marks: by subject, by question type, by chapter and by question, with every high-impact question cropped from the paper.
- **Institute vs our top 10** (`…_Institute_vs_Our_Top10.pdf`) – how our top 10 compare with the institute's top 10 in the same program, question by question.

CLASS	EXAM	REPORT	FILE NAME	PAGES
G8	JEE Mains	Impact report	`G8_Mains_Top10_Impact_Report.pdf`	13
G8	JEE Mains	Institute vs our top 10	`G8_Mains_Institute_vs_Our_Top10.pdf`	7
G8	JEE Advanced	Impact report	`G8_Advance_Top10_Impact_Report.pdf`	9
G8	JEE Advanced	Institute vs our top 10	`G8_Advance_Institute_vs_Our_Top10.pdf`	7
G9	JEE Mains	Impact report	`G9_Mains_Top10_Impact_Report.pdf`	12
G9	JEE Mains	Institute vs our top 10	`G9_Mains_Institute_vs_Our_Top10.pdf`	7
G9	JEE Advanced	Impact report	`G9_Advance_Top10_Impact_Report.pdf`	13
G9	JEE Advanced	Institute vs our top 10	`G9_Advance_Institute_vs_Our_Top10.pdf`	5
G10	JEE Mains	Impact report	`G10_Mains_Top10_Impact_Report.pdf`	9
G10	JEE Mains	Institute vs our top 10	`G10_Mains_Institute_vs_Our_Top10.pdf`	1
G10	JEE Advanced	Impact report	`G10_Advance_Top10_Impact_Report.pdf`	10
G10	JEE Advanced	Institute vs our top 10	`G10_Advance_Institute_vs_Our_Top10.pdf`	6

A2 Who is analysed

Our top 10 = the 10 students of this campus with the best **program rank** in that test (program rank = rank within the batch, e.g. Pragya G8 (EXI) Achievers, across all campuses; see *Program rank* in the glossary). If two or more students tie at 10th place, everyone tied is included – so a group can have 11 students. This happened once: Class 10 JEE Mains has **11** students (two tied at program rank 10). All "out of 10" wording then becomes "out of 11".

CLASS 8 Mains top 10 = program ranks 11, 17, 26, 29, 40, 45, 51, 55, 56, 57. Advanced top 10 = program ranks 14, 17, 27, 29, 38, 52, 54, 58, 59, 64 (a different set of students, because ranks differ between the two tests).

A3 Right, wrong, skipped – and "lost"

For every student and every question, the final marks sheet says one of three things:

MARK	MEANING	SHOWN IN REPORTS AS
R – right	answered correctly	✓ on a light-green cell; green part of the type bars
W – wrong	answered, but incorrectly (negative marks apply where the paper has them)	✗ on a red cell; red bars
U – unanswered / skipped	left blank	– on a grey cell; grey bars

A question is **"lost"** by a student if it is **W or U** – they did not get the marks. "Lost by 7/10" means 7 of the top 10 got it wrong or skipped it. "Lost" never means "lost the paper".

A4 Paper patterns and marking

The reports use the paper's own marking for every number. "Cost of a wrong answer" is what a student gives away compared with answering correctly: the marks not earned *plus* the negative mark.

JEE Mains (Classes 8, 9, 10) – 75 questions, 300 marks

SUBJECT	SECTION	QUESTIONS	CORRECT	WRONG	COST OF A WRONG ANSWER	COST OF A SKIP
Maths	Single correct	Q1–20	+4	−1	5 marks	4 marks
Maths	Numerical	Q21–25	+4	−1	5 marks	4 marks
Physics	Single correct	Q26–45	+4	−1	5 marks	4 marks
Physics	Numerical	Q46–50	+4	−1	5 marks	4 marks
Chemistry	Single correct	Q51–70	+4	−1	5 marks	4 marks
Chemistry	Numerical	Q71–75	+4	−1	5 marks	4 marks

JEE Advanced (Paper 2 pattern) – 51 questions, 180 marks

SUBJECT	SECTION	QUESTIONS	CORRECT	WRONG	COST OF A WRONG ANSWER	COST OF A SKIP
Maths	Single correct	Q1–4	+3	−1	4 marks	3 marks
Maths	Multiple correct	Q5–7	+4	−2	6 marks	4 marks
Maths	Integer	Q8–13	+4	0	4 marks	4 marks
Maths	Paragraph	Q14–17	+3	0	3 marks	3 marks
Physics	Single correct	Q18–21	+3	−1	4 marks	3 marks
Physics	Multiple correct	Q22–24	+4	−2	6 marks	4 marks
Physics	Integer	Q25–30	+4	0	4 marks	4 marks
Physics	Paragraph	Q31–34	+3	0	3 marks	3 marks
Chemistry	Single correct	Q35–38	+3	−1	4 marks	3 marks
Chemistry	Multiple correct	Q39–41	+4	−2	6 marks	4 marks
Chemistry	Integer	Q42–47	+4	0	4 marks	4 marks
Chemistry	Paragraph	Q48–51	+3	0	3 marks	3 marks

Multiple-correct questions give partial marks in the real paper (1–3 marks when only some of the correct options are chosen). The marks sheet only says right / wrong / skipped, so the reports treat "right" as full marks and "wrong" as −2. That is why Advanced marks figures carry a ≈ sign. The *counts* (how many students lost a question) are always exact.

Chemistry Section I in the Class 8 Advanced paper: the cover page says Q35–37, but the paper itself prints Q35–38 under Section I. The reports follow the paper.

A5 Where the data comes from

SOURCE	WHERE IT IS	WHAT IS TAKEN FROM IT	WHERE YOU SEE IT
Final marks sheets ("Toppers Vs Individual Student Qwise RWU Analysis", Excel)	`FINAL MARKS/MAINS MARKS`, `FINAL MARKS/ADV MARKS` — one file per batch and exam	Every student's R / W / U on every question; their program rank (the column is labelled AIR or ARI in the sheet) and total; the institute's *Top 10 / 20 / 50 / 100* rows	Everything about our top 10; the institute side of the comparison
Campus result sheets (Top10 Lost / Unanswered, PDF)	`MAINS REPORTS`, `ADV REPORTS`	The batch name (e.g. "Pragya G8 (EXI) Achievers") and campus; used as a cross-check of the marks sheets	Report titles
Final answer keys	Key sheets shared for Class 8; written into the tool's settings (`paper_config.py`)	Which questions are DEL (deleted) or revised	"del" markers; questions left out of every count
Question papers (PDF)	`QUESTION PAPERS/MAINS`, `QUESTION PAPERS/ADVANCE`	The picture of each question (with its paragraph, where there is one)	Question cards
All India Marks Analysis (Excel)	`PART TEST-1 ANALYSIS-COMMON/…COMMON_RANKING`	Each student's All-India rank (AIR) in the whole class programme	AIR columns
Mains & Advance averages (Excel)	`HYD_Nallakunta DD Colony_MAINS_ADV_AVERAGES (1).xlsx`, sheets `G8_ACH`, `G9_ACH`, `G10_ACH`	Each student's average over the weekly tests (WTM out of 300; WTA out of 240, scaled to 180) and how many they sat	Avg columns
Chapter tags and concept notes	Written by reading every question; stored in `paper_config.py`	Chapter of every question; a one-line "what it tests" for questions that appear in tables or cards	Chapter tables, question lists, cards

A6 How a report is made

1 · Read final marks sheet, batch name, AIR, averages › **2 · Final key** drop DEL questions (and any with no data) › **3 · Count & score** lost / wrong / skipped, marks lost, gaps › **4 · Cut questions** crop each needed question from the paper › **5 · Lay out** tables, charts, sentences › **6 · Print** PDF, checked automatically

All sentences in the reports (headlines, takeaways, notes) are generated from rules, not typed by hand, so they always match the numbers. The rules are listed in B3, C3 and Part D.

A7 How the final key is applied

```
Rule 1  DEL in the final key            → left out everywhere (“del”)
Rule 2  any other question              → counted as marked in the marks sheet
Rule 3  not DEL, but no R/W/U for anyone → cannot be counted (“n/a”)
```

Left-out questions also come off each subject's maximum, so a subject can be “out of 96” or “out of 50” instead of 100 or 60 (see *Subject maximum*).

EXAM	QUESTION	FINAL KEY SAYS	WHAT THE REPORTS DO
Class 8 Mains	Q20	DEL	Left out – “del” (Rule 1)
Class 8 Mains	Q49	answer 298	Counted as marked in the final marks sheet (Rule 2)
Class 8 Mains	Q67	answer D	Counted as marked in the final marks sheet (Rule 2)
Class 8 Advanced	Q14	DEL	Left out – “del” (Rule 1)
Class 8 Advanced	Q17	DEL	Left out – “del” (Rule 1)
Class 8 Advanced	Q50	DEL	Left out – “del” (Rule 1)
Class 8 Advanced	Q13	answer 4046	No student has an entry in the sheet → left out – “n/a” (Rule 3)
Class 8 Advanced	Q29	answer 9	No student has an entry in the sheet → left out – “n/a” (Rule 3)
Class 8 Advanced	Q25	answer 1 OR 2	Counted as marked in the final marks sheet (Rule 2)
Class 8 Advanced	Q48	answer 2 OR 5	Counted as marked in the final marks sheet (Rule 2)
Class 8 Advanced	Q51	answer 3	Counted as marked in the final marks sheet (Rule 2)

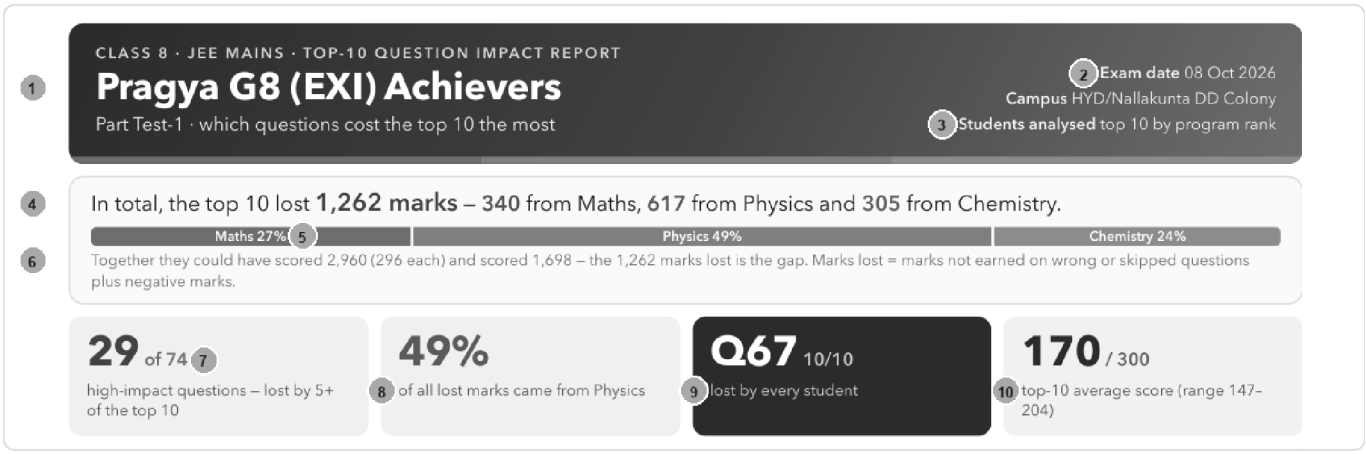
> **Classes 9 and 10**
>
> No final key was supplied, so only questions with no entry for anyone in the marks sheet are left out: Class 9 Mains Q58 and Advanced Q45; Class 10 Mains Q56 and Advanced Q25. Sending their final keys would let the same rules be applied.

PART B

The impact report, section by section

Worked on `G8_Mains_Top10_Impact_Report.pdf` (Pragya G8 (EXI) Achievers, JEE Mains, 08 Oct 2026). Each screenshot is the real report; the numbered list underneath explains each marker. B11 covers what changes in the Advanced report.

B1 Header, headline and the four tiles



Top of page 1 – G8 Mains impact report

1. **Batch name.** The program batch the students belong to, taken from the campus result sheet. The small line above it gives class, exam and report type; the line below gives the test.

2. **Exam date and campus.** From the final marks sheet and result sheet.

3. **Students analysed.** Always "top 10 by program rank", or "11 students (tie at 10th place)" when there is a tie. See A2.

4. **Headline – total marks lost.** All marks the top 10 together did not get, split by subject. It adds up exactly: 340 + 617 + 305 = 1,262.

   How: for every question and every student who lost it: wrong → the question's marks + its negative mark; skipped → the question's marks. Add everything up.

5. **Subject bar.** Each subject's share of the total, drawn to scale and labelled when wide enough: Maths 27%, Physics 49%, Chemistry 24%.

6. **Tally line** (Mains only, because Mains marking is exact). Proves the headline from the official totals: the top 10 could have scored 10 × 296 = 2,960; their official totals add up to 1,698; 2,960 − 1,698 = 1,262. (296 = 300 minus the deleted Q20's 4 marks.)

7. **High-impact questions.** How many valid questions were lost by 5 or more of the top 10 – 29 of 74. Deleted questions are not counted in the "of".

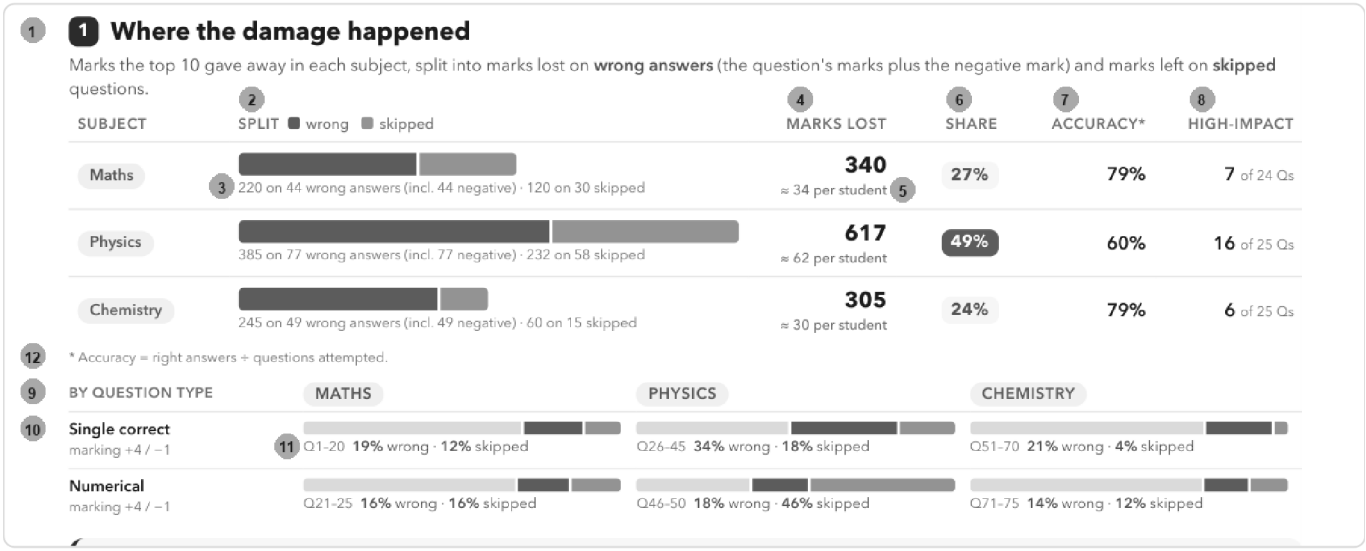
8. **Worst subject's share.** The subject with the most marks lost and its share of the total (49% – Physics). The tile takes that subject's colour.

9. **Most-lost question.** The question lost by the most students (ties broken by most wrong answers). "Lost by every student" when it is 10/10; "plus N more questions at 10/10" when several share the top spot.

   CLASS 8 Q67, 10/10 – all 10 of the top 10 answered it wrong.

10 **Top-10 average score.** The average of the 10 official totals, with the lowest and highest in brackets: 170 / 300 (range 147–204).

B2 Section 1 — Where the damage happened



Section 1 — subject table and question-type table

1 **Section title.** Section 1 answers "which subject cost us the most?".

2 **Split bar.** Red part = marks lost on wrong answers; grey part = marks lost on skipped questions. All three bars share one scale (the biggest subject fills the width), so lengths can be compared across subjects.

3 **The detail line under each bar** spells the split out.

CLASS 8 Maths: "220 on 44 wrong answers (incl. 44 negative) · 120 on 30 skipped" → 44 × 5 = 220 (of which 44 × 1 = 44 are negative marks), and 30 × 4 = 120.

4 **Marks lost.** The subject total: wrong-answer marks + skipped marks (220 + 120 = 340 for Maths).

5 **Per student.** Marks lost ÷ number of students, rounded: 340 ÷ 10 ≈ 34.

6 **Share.** Subject marks lost ÷ all marks lost. The largest share is shown in a red pill.

CLASS 8 Physics 617 ÷ 1,262 = 49%.

7 **Accuracy*.** Of the answers the students actually attempted, how many were right: right ÷ (right + wrong). Skipped questions do not count either way.

CLASS 8 Physics: 25 questions × 10 students = 250 answers; 77 wrong and 58 skipped leave 115 right; 115 ÷ (115 + 77) = 60%.

8 **High-impact.** How many of the subject's valid questions are high-impact (lost by 5+): Maths 7 of 24 (24, not 25, because Q20 is deleted).

9 **By question type.** The same picture for each section of the paper, side by side for the three subjects.

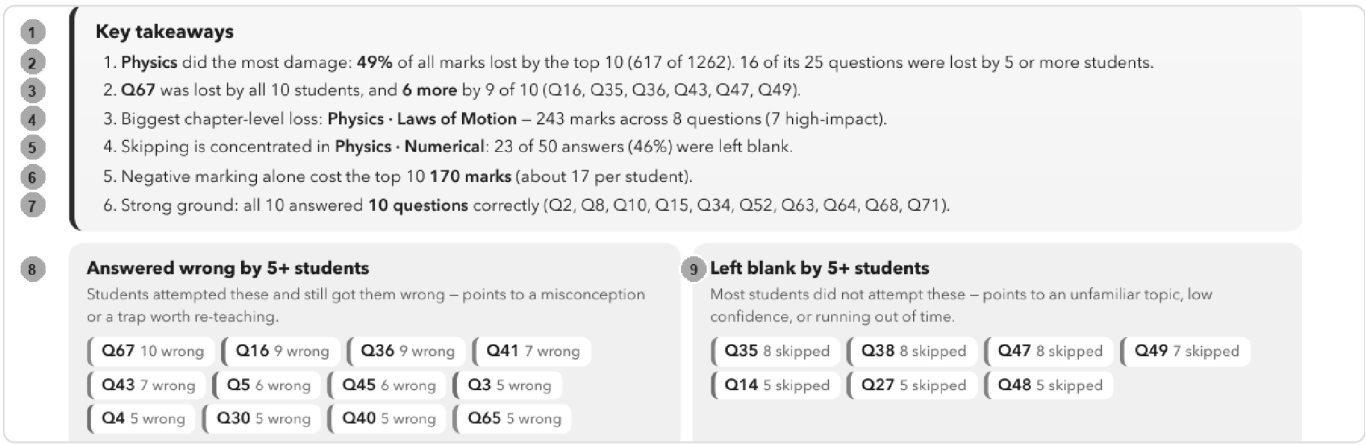
10 **Type name and its marking** (e.g. Single correct, +4 / −1).

11 **Type bar and percentages.** Green = right, red = wrong, grey = skipped, as shares of all answers in that block (questions × 10 students). The question range is shown first.

CLASS 8 Maths Single correct (Q1–20, Q20 deleted → 19 questions, 190 answers): 19% wrong, 12% skipped.

12 **Footnote** defining accuracy.

B3 Key takeaways and the two diagnosis boxes



Bottom of page 1 – takeaways and diagnosis

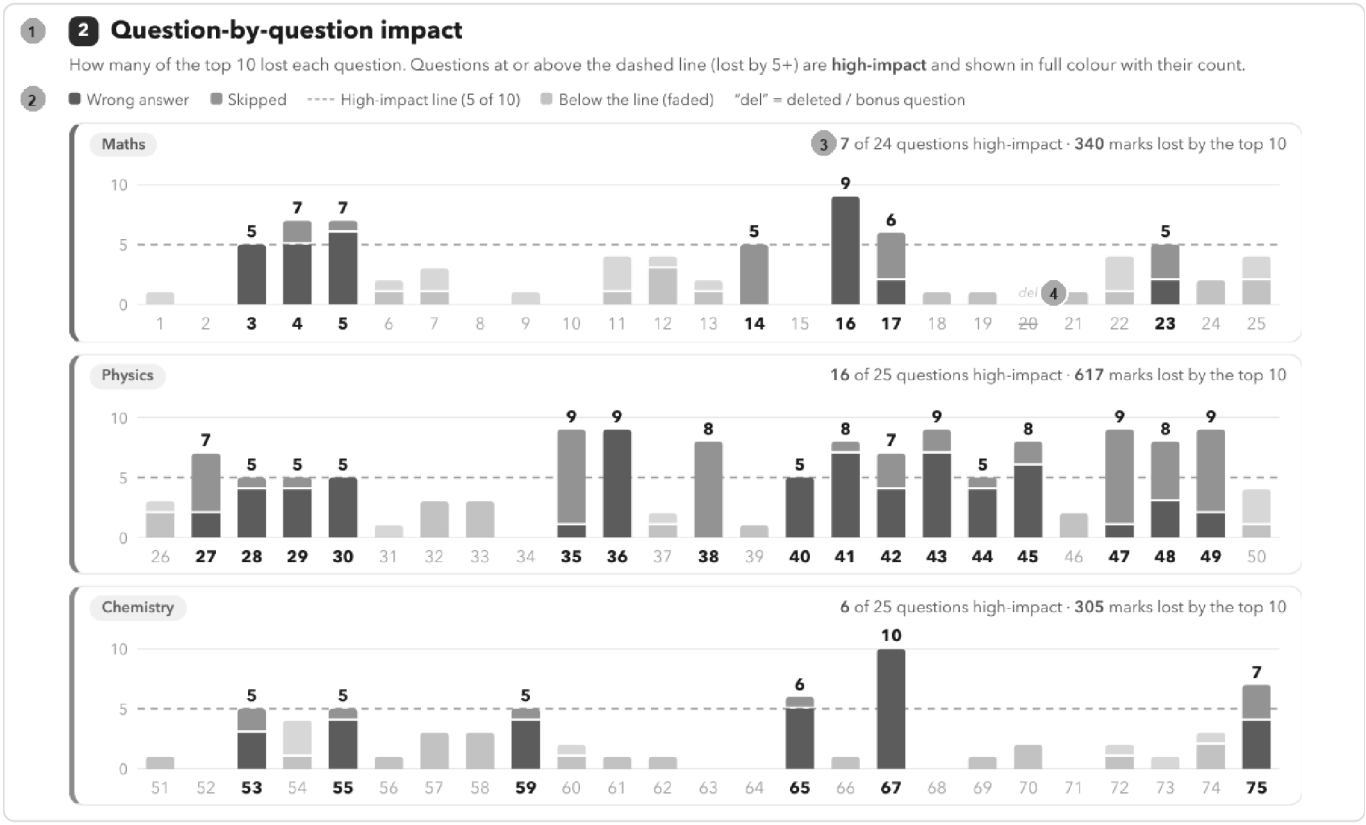
The takeaways are written automatically. Each sentence has a rule and appears only when the rule finds something worth saying, so different reports show different lines.

1. **Key takeaways.** Up to seven sentences, always in this order.
2. **"…did the most damage"** – always shown. The subject with the most marks lost, its share, and how many of its questions were high-impact.
3. **"…lost by all 10 students"** – shown if any question was lost by everyone; also lists those lost by 9 of 10. (If none was lost by all 10 but some by 9, it says so instead.)
4. **"Biggest chapter-level loss"** – always shown. The chapter with the most marks lost (the top row of section 3).
5. **"Skipping is concentrated in…"** – the section of the paper with the highest share of blank answers, shown only if that share is **15% or more**.

   CLASS 8 Physics · Numerical: 23 of 50 answers (46%) blank.
6. **"Negative marking alone cost…"** – the total of negative marks over all wrong answers, and per student: 44 + 77 + 49 = 170.
7. **"Strong ground"** – questions every one of the top 10 got right.
8. **Answered wrong by 5+ students.** Questions with 5 or more *wrong* answers, most first. Students tried these and still got them wrong – the sign of a misconception or a trap worth re-teaching. The coloured edge of each pill is the subject.
9. **Left blank by 5+ students.** Questions with 5 or more *skips*. Most students did not even try – the sign of an untaught or unfamiliar topic, low confidence, or running out of time.

> **A takeaway you will see in Advanced but not here:** "*Paragraph questions were the weakest type…*". It appears only when one question type is lost at least **1.5 times** as often as the paper overall. Class 8 Advanced: "**Paragraph** questions were the weakest type in the paper: **77%** of answers wrong or blank, against 36% across all questions."

B4 Section 2 — Question-by-question impact



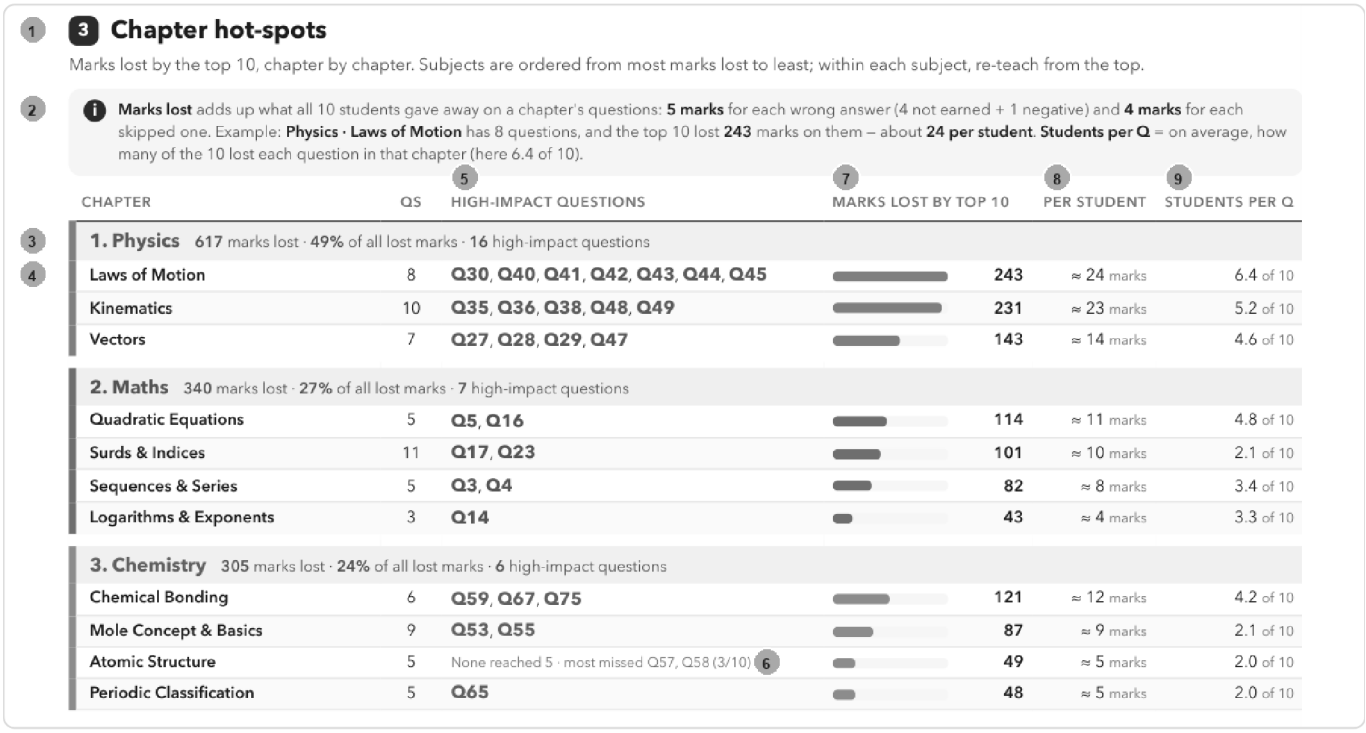
Section 2 — one panel per subject (Maths panel shown in full)

1. **Section title.** One bar per question, one panel per subject.
2. **Legend.** ■ red = wrong answers, ■ grey = skipped, dashed line = the high-impact line (5 of 10), pale bars = below the line, "del" = a deleted question.
3. **Panel summary.** High-impact questions in this subject (of its valid questions) and the subject's marks lost.
4. **"del"** marks a question left out by the final key; its number is struck through (here Q20).

How to read a bar. Height = how many of the top 10 lost the question (0 to 10). The red part at the bottom is the number who answered wrong; the grey part on top is the number who skipped. A bar that reaches the dashed line is high-impact: it is drawn in full colour with its count on top and its question number in bold. Bars below the line are faded.

CLASS 8 Q16 is a red bar of 9 (9 wrong). Q14 is a grey bar of 5 (5 skipped). Q23 stacks 2 wrong + 3 skipped = 5.

B5 Section 3 – Chapter hot-spots



3 Chapter hot-spots

Marks lost by the top 10, chapter by chapter. Subjects are ordered from most marks lost to least; within each subject, re-teach from the top.

Marks lost adds up what all 10 students gave away on a chapter's questions: **5 marks** for each wrong answer (4 not earned + 1 negative) and **4 marks** for each skipped one. Example: **Physics · Laws of Motion** has 8 questions, and the top 10 lost **243** marks on them – about **24 per student**. **Students per Q** = on average, how many of the 10 lost each question in that chapter (here 6.4 of 10).

CHAPTER	QS	HIGH-IMPACT QUESTIONS	MARKS LOST BY TOP 10	PER STUDENT	STUDENTS PER Q
1. Physics 617 marks lost · 49% of all lost marks · 16 high-impact questions					
Laws of Motion	8	Q30, Q40, Q41, Q42, Q43, Q44, Q45	243	≈ 24 marks	6.4 of 10
Kinematics	10	Q35, Q36, Q38, Q48, Q49	231	≈ 23 marks	5.2 of 10
Vectors	7	Q27, Q28, Q29, Q47	143	≈ 14 marks	4.6 of 10
2. Maths 340 marks lost · 27% of all lost marks · 7 high-impact questions					
Quadratic Equations	5	Q5, Q16	114	≈ 11 marks	4.8 of 10
Surds & Indices	11	Q17, Q23	101	≈ 10 marks	2.1 of 10
Sequences & Series	5	Q3, Q4	82	≈ 8 marks	3.4 of 10
Logarithms & Exponents	3	Q14	43	≈ 4 marks	3.3 of 10
3. Chemistry 305 marks lost · 24% of all lost marks · 6 high-impact questions					
Chemical Bonding	6	Q59, Q67, Q75	121	≈ 12 marks	4.2 of 10
Mole Concept & Basics	9	Q53, Q55	87	≈ 9 marks	2.1 of 10
Atomic Structure	5	None reached 5 · most missed Q57, Q58 (3/10)	49	≈ 5 marks	2.0 of 10
Periodic Classification	5	Q65	48	≈ 5 marks	2.0 of 10

Section 3 – chapters grouped by subject

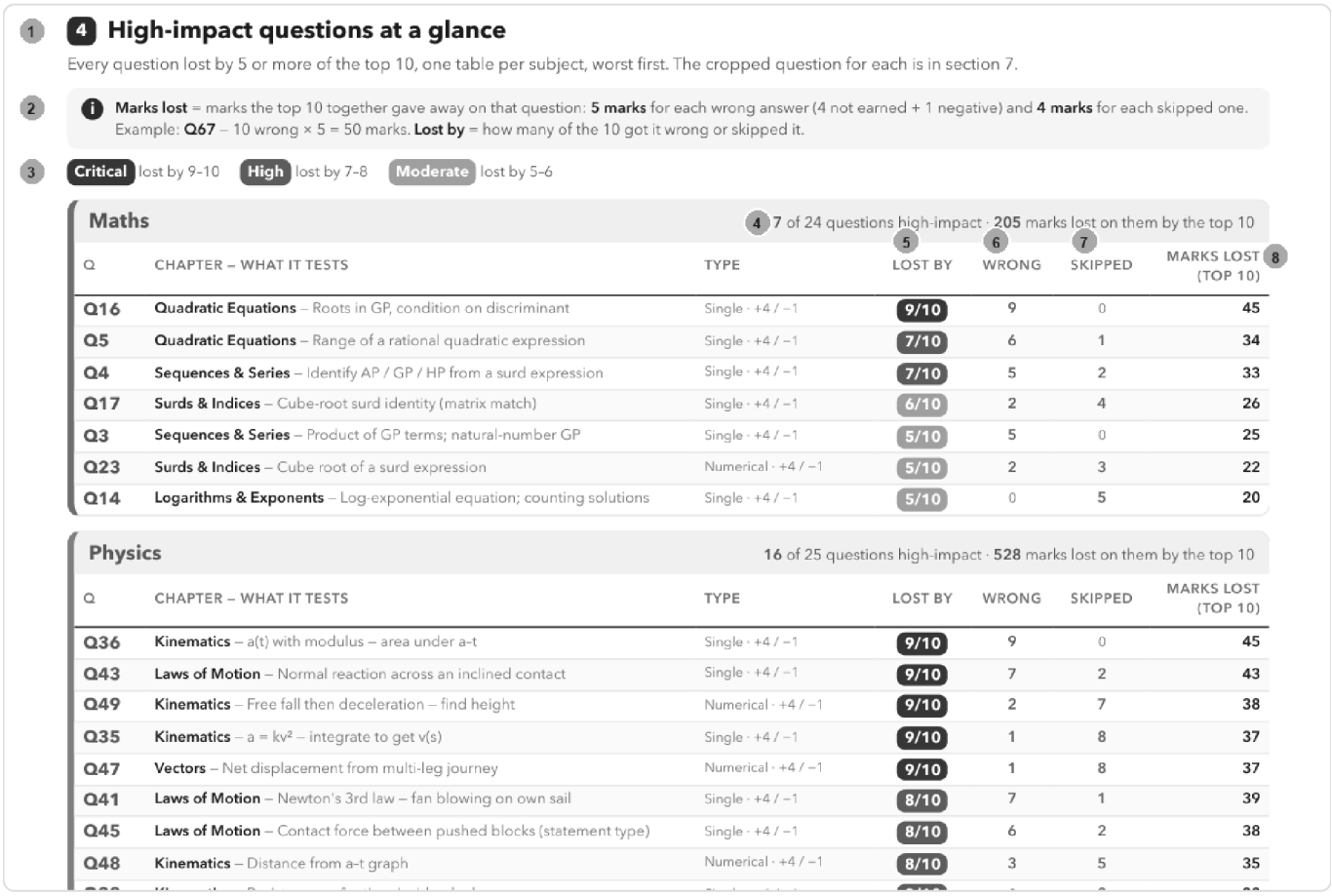
1. **Section title.** Where to re-teach first, chapter by chapter.
2. **Info box.** Restates the marks-lost rule with a live example from this report.
3. **Subject block.** Subjects in order of marks lost (most first), each with its marks lost, share of all lost marks, and number of high-impact questions.
4. **Chapter row.** Within a subject, chapters run from most marks lost to least.
5. **High-impact questions** in that chapter.
6. **"None reached 5 · most missed …"** – the chapter has no high-impact question; it names the most-missed question(s) instead and how many lost them (or says "nobody lost these").

   CLASS 8 Atomic Structure: none of its 5 questions was lost by 5+; the worst were Q57, Q58 (3/10).
7. **Marks lost by top 10.** The sum of marks lost on the chapter's questions. One bar scale for the whole table, so chapters can be compared across subjects.
8. **Per student.** Chapter marks lost ÷ 10.
9. **Students per Q.** On average, how many of the 10 lost each question in the chapter (the average of the "lost by" counts).

   CLASS 8 Physics · Laws of Motion: 8 questions, 243 marks lost, ≈ 24 per student, 6.4 of 10 per question.

Chapter tags were assigned by reading every question in the paper. If your department files a question under a different chapter, the tag can be changed in `paper_config.py` and the reports rebuilt (F6).

B6 Section 4 – High-impact questions at a glance



1 **4 High-impact questions at a glance**

Every question lost by 5 or more of the top 10, one table per subject, worst first. The cropped question for each is in section 7.

2 **Marks lost** = marks the top 10 together gave away on that question: **5 marks** for each wrong answer (4 not earned + 1 negative) and **4 marks** for each skipped one. Example: **Q67** – 10 wrong × 5 = 50 marks. **Lost by** = how many of the 10 got it wrong or skipped it.

3 **Critical** lost by 9–10 **High** lost by 7–8 **Moderate** lost by 5–6

Maths — 4 7 of 24 questions high-impact · **205** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	5 LOST BY	6 WRONG	7 SKIPPED	MARKS LOST (TOP 10) 8
Q16	**Quadratic Equations** – Roots in GP, condition on discriminant	Single · +4 / −1	9/10	9	0	45
Q5	**Quadratic Equations** – Range of a rational quadratic expression	Single · +4 / −1	7/10	6	1	34
Q4	**Sequences & Series** – Identify AP / GP / HP from a surd expression	Single · +4 / −1	7/10	5	2	33
Q17	**Surds & Indices** – Cube-root surd identity (matrix match)	Single · +4 / −1	6/10	2	4	26
Q3	**Sequences & Series** – Product of GP terms; natural-number GP	Single · +4 / −1	5/10	5	0	25
Q23	**Surds & Indices** – Cube root of a surd expression	Numerical · +4 / −1	5/10	2	3	22
Q14	**Logarithms & Exponents** – Log-exponential equation; counting solutions	Single · +4 / −1	5/10	0	5	20

Physics — **16** of 25 questions high-impact · **528** marks lost on them by the top 10

Q	CHAPTER – WHAT IT TESTS	TYPE	LOST BY	WRONG	SKIPPED	MARKS LOST (TOP 10)
Q36	**Kinematics** – a(t) with modulus – area under a-t	Single · +4 / −1	9/10	9	0	45
Q43	**Laws of Motion** – Normal reaction across an inclined contact	Single · +4 / −1	9/10	7	2	43
Q49	**Kinematics** – Free fall then deceleration – find height	Numerical · +4 / −1	9/10	2	7	38
Q35	**Kinematics** – a = kv² – integrate to get v(s)	Single · +4 / −1	9/10	1	8	37
Q47	**Vectors** – Net displacement from multi-leg journey	Numerical · +4 / −1	9/10	1	8	37
Q41	**Laws of Motion** – Newton's 3rd law – fan blowing on own sail	Single · +4 / −1	8/10	7	1	39
Q45	**Laws of Motion** – Contact force between pushed blocks (statement type)	Single · +4 / −1	8/10	6	2	38
Q48	**Kinematics** – Distance from a-t graph	Numerical · +4 / −1	8/10	3	5	35

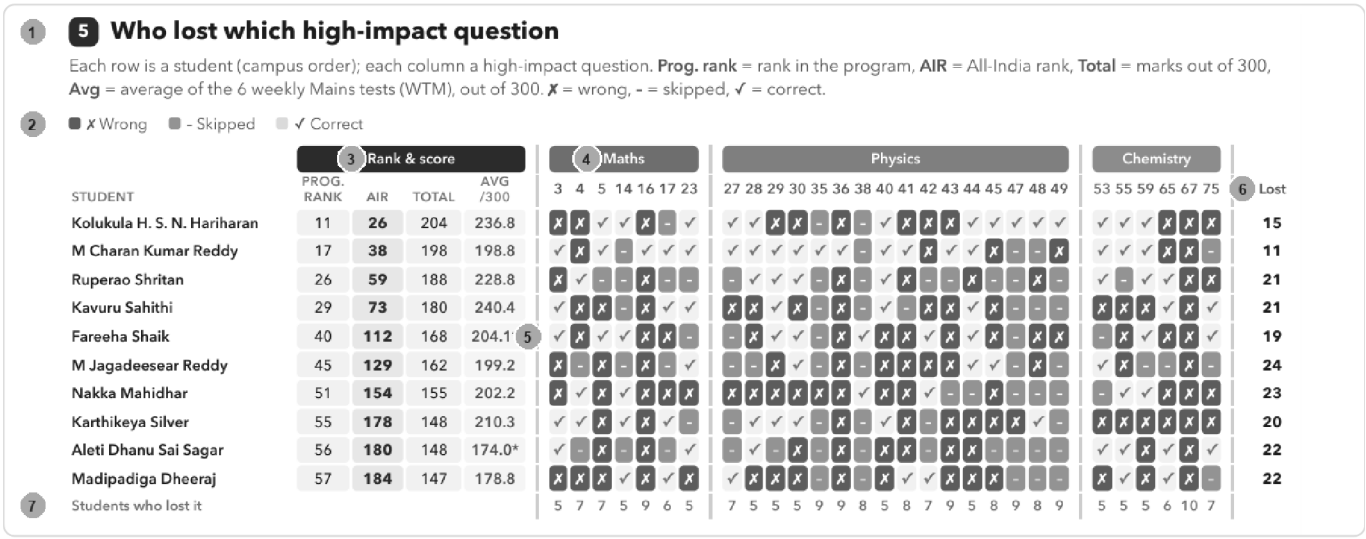
Section 4 – one table per subject (top of page shown)

1. **Section title.** The to-do list: every high-impact question, one table per subject, worst first.
2. **Info box.** The marks-lost rule with an example: Q67 → 10 wrong × 5 = 50.
3. **Severity legend.** The colour of each "Lost by" pill: **Critical** (dark red) = lost by 9–10 of the top 10, **High** (red) = 7–8, **Moderate** (light red) = 5–6.
4. **Subject header.** How many of the subject's questions are high-impact and the marks lost on them.
5. **Lost by.** Students (of 10) who got it wrong or skipped it, in a pill coloured by level.
6. **Wrong.** How many answered wrong.
7. **Skipped.** How many left it blank. Wrong + Skipped = Lost by.
8. **Marks lost (top 10).** wrong × (marks + negative) + skipped × marks.

   CLASS 8 Q49 (Numerical, +4 / −1): 2 wrong × 5 + 7 skipped × 4 = 38.

The other columns: **Q** (question number in the paper), **Chapter – what it tests** (chapter tag plus a one-line concept note), **Type** (section of the paper and its marking). Rows are ordered by Lost by, then by Wrong (more wrong answers first), then by question number.

B7 Section 5 — Who lost which high-impact question



5 Who lost which high-impact question

Each row is a student (campus order); each column a high-impact question. **Prog. rank** = rank in the program, **AIR** = All-India rank, **Total** = marks out of 300, **Avg** = average of the 6 weekly Mains tests (WTM), out of 300. ✗ = wrong, - = skipped, ✓ = correct.

■ ✗ Wrong ■ - Skipped ■ ✓ Correct

	Rank & score				Maths						Physics										Chemistry							
STUDENT	PROG. RANK	AIR	TOTAL	AVG /300	3	4	5	14	16	17	23	27	28	29	30	35	36	38	40	41	42	43	44	45	47	48	49	53	55	59	65	67	75	Lost
Kolukula H. S. N. Hariharan	11	26	204	236.8	✗	✗	✓	✓	✗	-	✓	✓	✓	✗	✗	-	✗	-	✓	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗	15
M Charan Kumar Reddy	17	38	198	198.8	✓	✗	✓	-	✓	✓	✓	✓	✓	✓	✓	✓	✓	-	✓	✓	✗	✓	✓	✗	-	-	✗	✓	✓	✓	✗	✗	-	11
Ruperao Shritan	26	59	188	228.8	✗	✓	-	-	✗	-	-	-	✓	✓	✓	-	✗	-	✓	✗	-	-	✗	-	-	✗	-	✓	-	✓	✓	✗	✗	21
Kavuru Sahithi	29	73	180	240.4	✓	✗	✗	-	✗	✓	✓	✗	✗	✓	✗	-	✗	-	✓	-	✗	✗	✓	✗	-	-	-	✗	✗	✗	✓	✗	✓	21
Fareeha Shaik	40	112	168	204.1*	✓	✗	✓	✓	✗	✗	-	-	✗	✓	✓	-	✗	✓	✗	✗	✓	✗	✓	✗	-	✗	✗	-	✗	✓	✗	✗	✓	19
M Jagadeesear Reddy	45	129	162	199.2	✗	-	✗	-	✗	-	✓	-	-	✗	✓	-	✗	-	✗	✗	✗	✗	✓	✓	-	✗	-	✓	✗	-	-	✗	-	24
Nakka Mahidhar	51	154	155	202.2	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓	-	-	✗	-	-	-	-	✓	✓	✗	✗	✗	23
Karthikeya Silver	55	178	148	210.3	✓	✓	✗	✓	✗	✓	-	-	✓	✓	✓	-	✗	-	✓	✗	-	✗	✗	✗	✗	✓	-	✗	✗	✗	✗	✗	✗	20
Aleti Dhanu Sai Sagar	56	180	148	174.0*	✓	-	✗	-	✗	-	✓	-	✓	-	✗	-	✗	✗	-	✗	-	✗	✗	-	-	-	-	✓	✓	✗	✓	✗	✓	22
Madipadiga Dheeraj	57	184	147	178.8	✗	✗	✗	✓	✗	✓	✗	✓	✗	✗	✗	-	✗	-	✗	✓	✓	✗	✗	✗	-	-	-	✗	✓	✗	✓	✗	-	22
Students who lost it					5	7	7	5	9	6	5	7	5	5	5	9	9	8	5	8	7	9	5	8	9	8	9	5	5	5	6	10	7	

Section 5 — student × question grid

1. **Section title.** One row per student (in program-rank order), one column per high-impact question.

2. **Legend.** ✗ on red = wrong, – on grey = skipped, ✓ on green = right.

3. **Rank & score** — four columns about the student:

   How: Prog. rank = rank in the batch across all campuses (from the final marks sheet) · AIR = All-India rank in the whole class programme (All India Marks Analysis) · Total = this test's official total (out of 300 Mains / 180 Advanced) · Avg = average over the weekly tests (WTM, out of 300, for Mains; WTA for Advanced — marked out of 240, scaled to 180 to match the paper).

   CLASS 8 Kolukula Hema Shiva Naga Hariharan: program rank 11, AIR 26, total 204, weekly average 236.8.

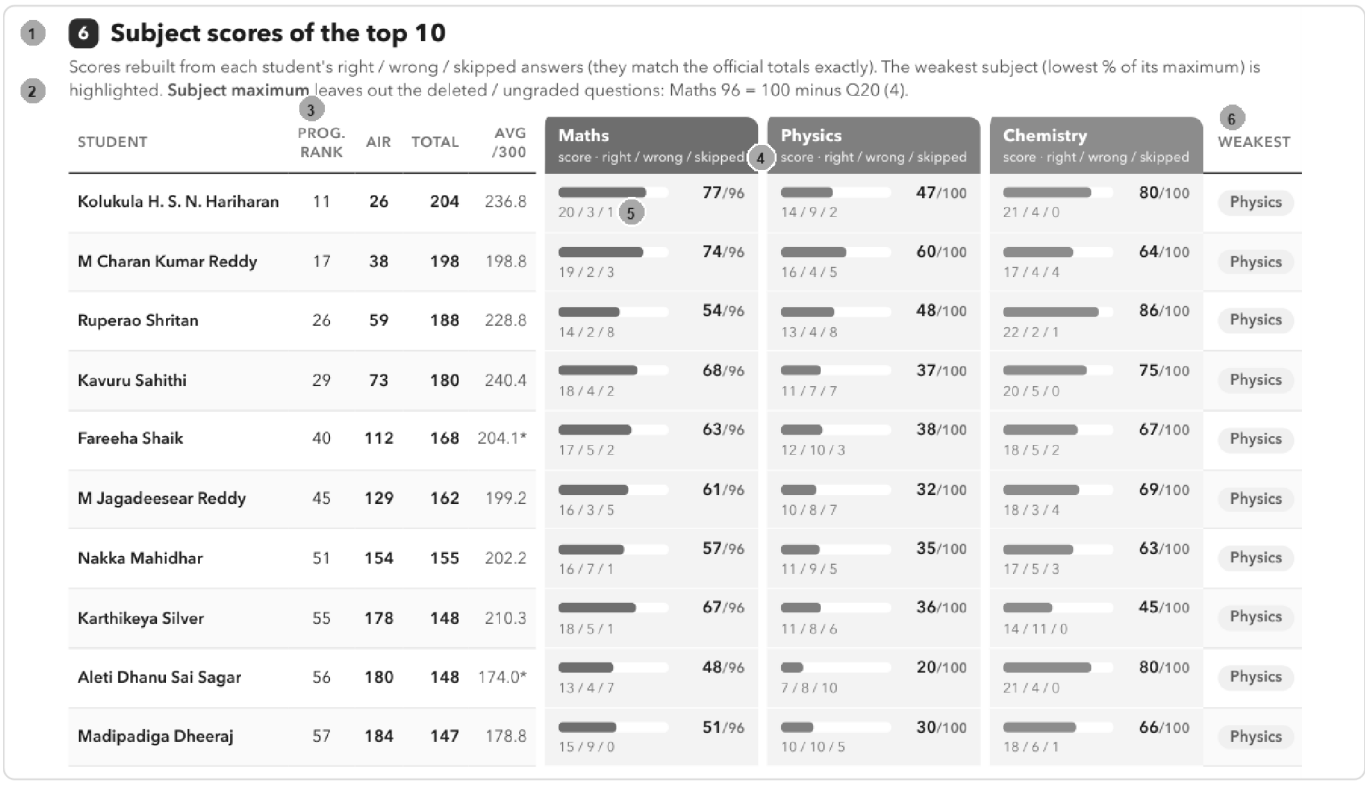
4. **Subject groups.** Columns are grouped Maths | Physics | Chemistry with a divider between groups; question numbers ascend within each group.

5. *** after an average** = the student sat fewer weekly tests than were held, so the average is over fewer tests. The notes list them: Fareeha Shaik (5/6), Aleti Dhanu Sai Sagar (3/6).

6. **Lost.** How many of the *high-impact* questions this student lost (not all questions in the paper).

7. **Students who lost it.** For each column, how many of the 10 lost that question — the same number as "Lost by" in section 4.

B8 Section 6 — Subject scores of the top 10



6 Subject scores of the top 10

Scores rebuilt from each student's right / wrong / skipped answers (they match the official totals exactly). The weakest subject (lowest % of its maximum) is highlighted. **Subject maximum** leaves out the deleted / ungraded questions: Maths 96 = 100 minus Q20 (4).

STUDENT	PROG. RANK	AIR	TOTAL	AVG /300	Maths score · right / wrong / skipped	Physics score · right / wrong / skipped	Chemistry score · right / wrong / skipped	WEAKEST
Kolukula H. S. N. Hariharan	11	26	204	236.8	77/96 20 / 3 / 1	47/100 14 / 9 / 2	80/100 21 / 4 / 0	Physics
M Charan Kumar Reddy	17	38	198	198.8	74/96 19 / 2 / 3	60/100 16 / 4 / 5	64/100 17 / 4 / 4	Physics
Ruperao Shritan	26	59	188	228.8	54/96 14 / 2 / 8	48/100 13 / 4 / 8	86/100 22 / 2 / 1	Physics
Kavuru Sahithi	29	73	180	240.4	68/96 18 / 4 / 2	37/100 11 / 7 / 7	75/100 20 / 5 / 0	Physics
Fareeha Shaik	40	112	168	204.1*	63/96 17 / 5 / 2	38/100 12 / 10 / 3	67/100 18 / 5 / 2	Physics
M Jagadeesear Reddy	45	129	162	199.2	61/96 16 / 3 / 5	32/100 10 / 8 / 7	69/100 18 / 3 / 4	Physics
Nakka Mahidhar	51	154	155	202.2	57/96 16 / 7 / 1	35/100 11 / 9 / 5	63/100 17 / 5 / 3	Physics
Karthikeya Silver	55	178	148	210.3	67/96 18 / 5 / 1	36/100 11 / 8 / 6	45/100 14 / 11 / 0	Physics
Aleti Dhanu Sai Sagar	56	180	148	174.0*	48/96 13 / 4 / 7	20/100 7 / 8 / 10	80/100 21 / 4 / 0	Physics
Madipadiga Dheeraj	57	184	147	178.8	51/96 15 / 9 / 0	30/100 10 / 10 / 5	66/100 18 / 6 / 1	Physics

Section 6 — each student's score per subject

1. **Section title and intro.** For Mains the rebuilt scores match the official totals exactly; for Advanced they are approximate (≈, see B11).

2. **Subject maximum.** The subject's marks on the paper minus any left-out question: Maths 96 = 100 − Q20 (4).

3. **Prog. rank, AIR, Total, Avg** — the same four columns as in section 5.

4. **Subject column header.** Each cell shows the score / maximum, a bar for the percentage, and below it right / wrong / skipped counts.

5. **Worked example.**

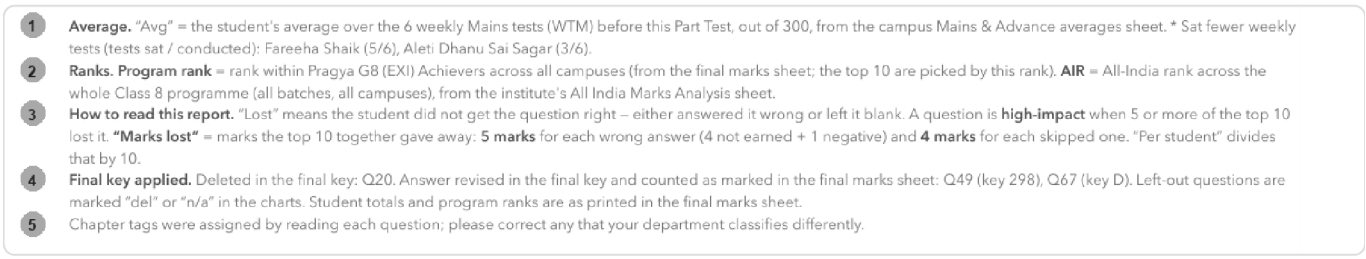
   CLASS 8 Kolukula Hema Shiva Naga Hariharan, Maths: 20 right / 3 wrong / 1 skipped → 96 − 3 × 5 − 1 × 4 = **77** / 96.

6. **Weakest.** The subject with the lowest percentage of its own maximum (so 47/100 is weaker than 77/96).

   CLASS 8 Kolukula Hema Shiva Naga Hariharan: Physics 47/100 = 47% is lowest → "Physics".

```
subject score = subject maximum − Σ(wrong × (marks + negative)) − Σ(skipped × marks)
check         Maths 77 + Physics 47 + Chemistry 80 = 204 = official total 204 ✓
```

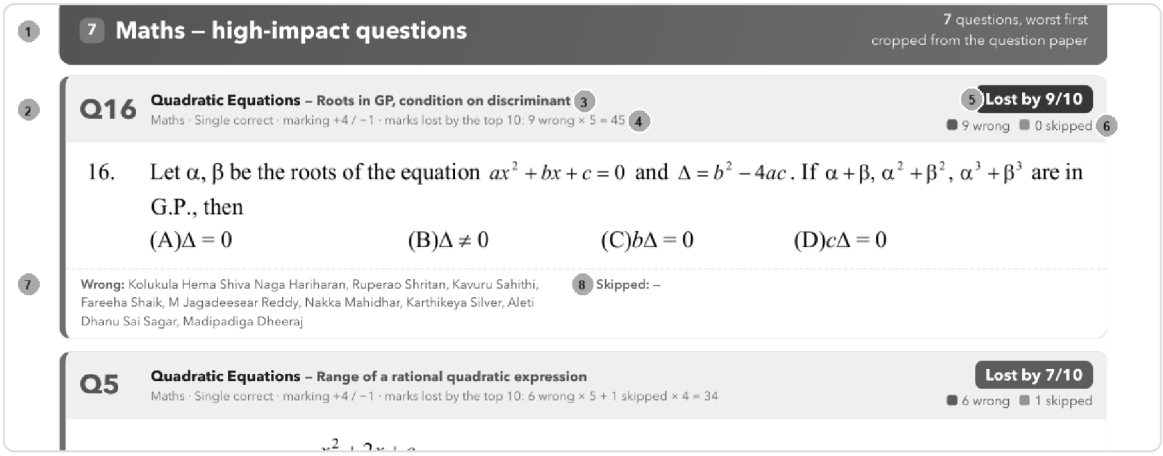
B9 The notes block



Notes at the bottom of section 6

1. **Average.** What "Avg" is, out of what, and who sat fewer weekly tests.
2. **Ranks.** The difference between program rank and AIR.
3. **How to read this report.** The definitions of "lost", "high-impact" and "marks lost".
4. **Final key applied.** Which questions the final key deleted or revised and how they were treated (A7).
5. **Chapter tags.** A reminder that chapters were assigned by reading the questions.

B10 Section 7 – Question cards



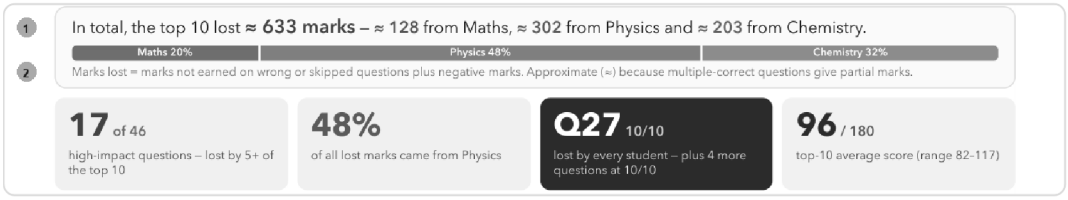
Section 7 – the first Maths card

1. **Subject banner.** Each subject starts on a new page so its cards can be handed to that subject's teacher. Cards run worst first (same order as section 4).
2. **Question number** in the paper.
3. **Chapter – what it tests.** The chapter tag and concept note.
4. **Detail line.** Subject · question type · marking · the marks-lost sum: Q16 → 9 wrong × 5 = 45.
5. **Lost by** pill, coloured Critical / High / Moderate.
6. **Wrong and skipped counts.**
7. **Wrong:** full names of the students who answered it wrong.
8. **Skipped:** full names of those who left it blank ("–" if nobody).

The question picture is cut automatically from the question paper (F4). For paragraph / passage questions the paragraph is included above the question.

B11 What is different in the Advanced report

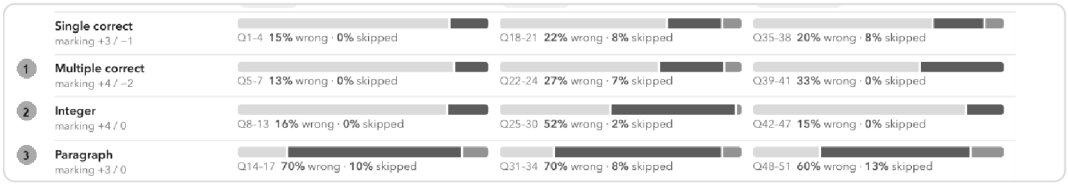
The Advanced report has exactly the same sections; only the points below differ.



G8 Advanced – headline

1. **≈ in front of marks.** Multiple-correct questions give partial marks, which the marks sheet cannot show, so every marks figure is approximate: about 633 marks lost (Maths ≈ 128, Physics ≈ 302, Chemistry ≈ 203). The tally line from B1 is not shown, because it would not add up exactly.

2. **The explanation of ≈** is printed right under the headline.

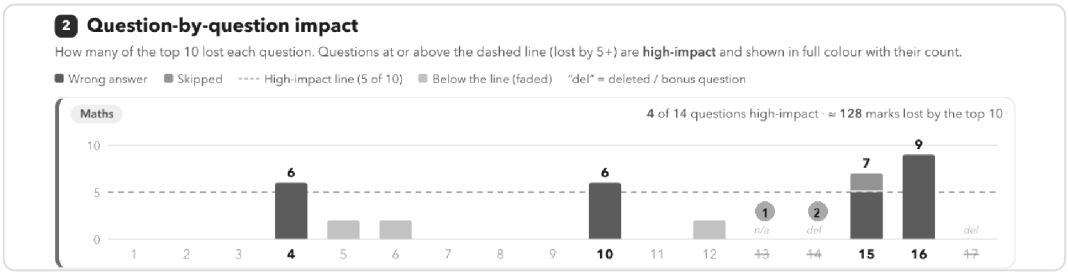


G8 Advanced – four question types

1. **Multiple correct** (+4 / −2): cost of a wrong answer 4 + 2 = 6.

2. **Integer** (+4 / 0): no negative mark, so a wrong answer costs 4, the same as a skip.

3. **Paragraph** (+3 / 0): cost 3 either way. In Class 8 this is the weakest type by far.



G8 Advanced – Maths panel of section 2

1. **"n/a"** – a question that is valid in the final key but has no right / wrong / skipped entry for anyone in the marks sheet (Q13 and Q29). It cannot be counted.

2. **"del"** – a question deleted in the final key (Q14, Q17, Q50).

- **Subject maxima shrink:** Maths 50 = 60 − Q13 (4) − Q14 (3) − Q17 (3); Physics 56 = 60 − Q29 (4); Chemistry 57 = 60 − Q50 (3).
- **Avg is scaled to 180.** The weekly Advanced tests (WTA) are marked out of 240 (80 per subject); the reports multiply each average by 180/240 so it sits on the same scale as this paper (60 per subject). Percentages are unchanged.
- **Section 6 scores (≈)** are rebuilt from right / wrong / skipped, which cannot show partial credit, so they do not match the official totals exactly. In Class 8 the rebuilt totals differ from the official ones by 0 to 9 marks (rebuilt − official).

PART C

The comparison report

Worked on `G8_Mains_Institute_vs_Our_Top10.pdf`. It answers one question: *how far behind the institute's best students are our best students, and on which questions?*

C1 What is being compared

GROUP	WHO	WHERE THE NUMBERS COME FROM
Institute top 10	The 10 best students in the whole program across all campuses — program ranks 1 to 10 (more if tied). Some of them may be our own students.	The *Top 10 Students* rows at the top of the final marks sheet: for each question, how many of them got it right, wrong or skipped.
Our top 10	This campus's top 10 by program rank (A2) — the same students as the impact report.	Each student's own row in the same sheet.

Both groups are scored the same way, on the same valid questions (final-key deletions are removed from both).

```
group marks on a question = marks × (number right) − negative × (number wrong)
group average score       = Σ group marks over all valid questions ÷ group size
gap per student           = institute average − our average   (positive = we are behind)
```

C2 Headline and the four tiles



Top of page 1 — G8 Mains comparison

1. **Headline.** Both averages, the gap per student, and how that gap splits by subject (only subjects where the institute is ahead are listed).
   CLASS 8 224.3 − 169.8 = 54.5 marks per student: Maths 3.0, Physics 30.7, Chemistry 20.8.
2. **Gap bar.** Each subject's share of the total gap (Maths 6%, Physics 56%, Chemistry 38%). Very small shares are drawn without a label.
3. **Method line.** How the scores were rebuilt. For Mains our rebuilt average equals the official one exactly (169.8), which proves the method; for Advanced it is approximate (≈).

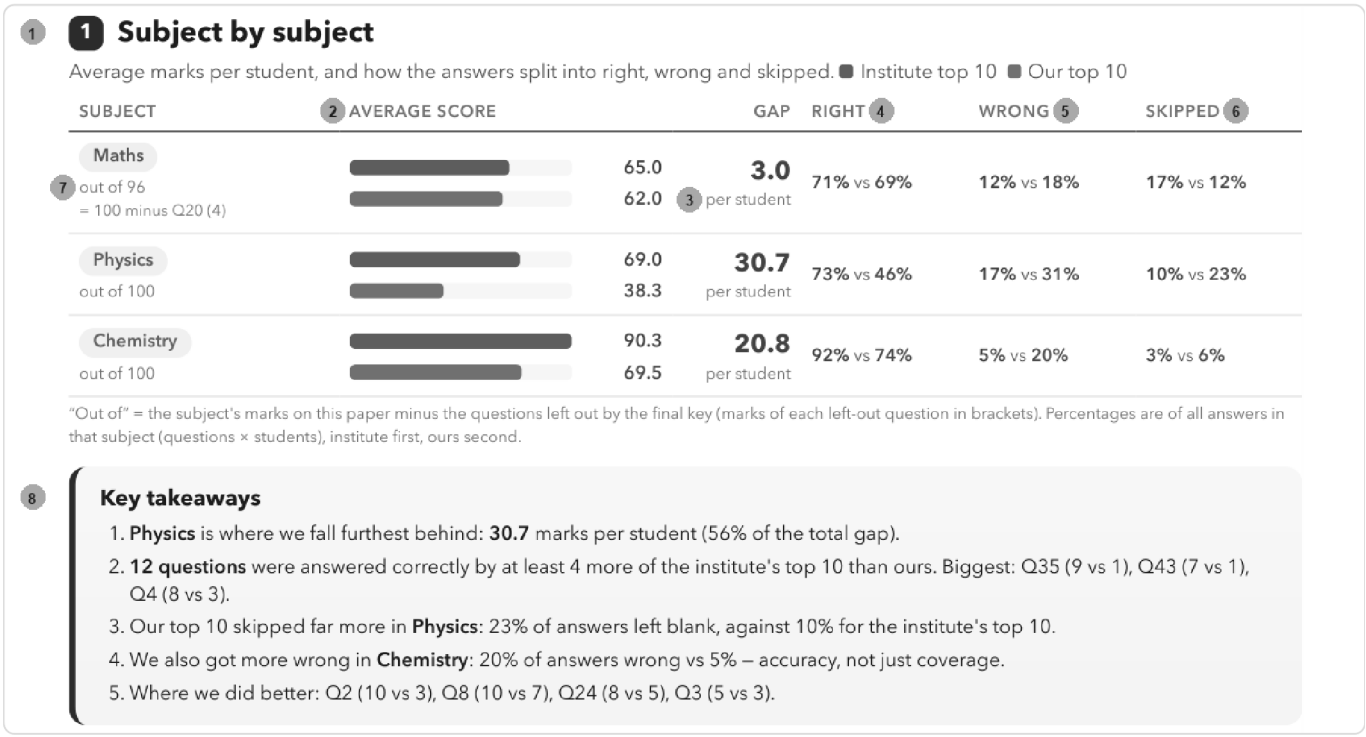
4 **Institute top 10 — average score**, out of the paper's valid maximum (296 = 300 − deleted Q20).

5 **Our top 10 — average score.**

6 **Gap questions.** Questions where at least 4 more of the institute's top 10 got the right answer than ours (the "gap line", which can drop to 2 — see C8).

7 **Ahead questions.** Questions where at least 2 more of *ours* got it right.

C3 Section 1 — Subject by subject



Section 1 — subject table and takeaways

1 **Section title.**

2 **Average score bars.** ■ violet = institute top 10, ■ magenta = our top 10, average marks per student in that subject.

3 **Gap per student** = institute average − our average. A negative gap (shown in green with a minus sign) means our top 10 are ahead in that subject.

4 **Right** — share of all answers in the subject that were right, institute first, ours second (answers = questions × students).

5 **Wrong** — the same for wrong answers. A big difference here means an accuracy problem.

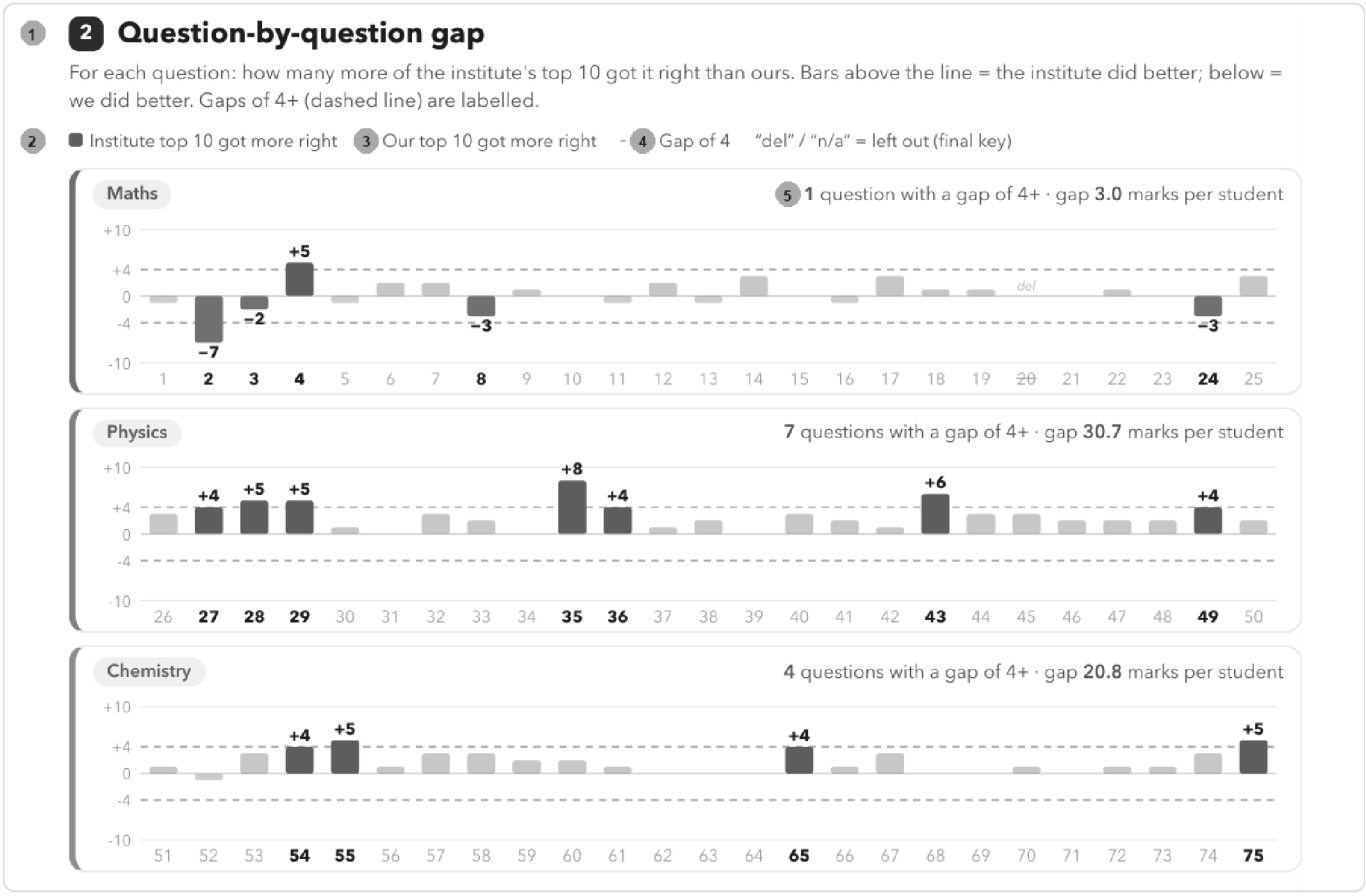
6 **Skipped** — the same for blanks. A big difference here means a coverage or confidence problem.

7 **Out of.** The subject's maximum after left-out questions, with the subtraction spelled out underneath ("= 100 minus Q20 (4)"). In Advanced: Maths 50 = 60 minus Q13 (4), Q14 (3), Q17 (3).

8 **Key takeaways** — generated by these rules, in this order:

How: (a) if some of the institute's top 10 are ours, say how many and what that means (C8); (b) the subject where we are furthest behind and its share of the gap; (c) how many gap questions there are and the three biggest; (d) a subject where we skip at least 5 percentage points more; (e) a subject where we answer wrong at least 5 points more; (f) any subject where we are ahead by more than 0.5 marks; (g) the questions where we did better.

C4 Section 2 — Question-by-question gap



Section 2 — one panel per subject

1 **Section title.**

2 **Violet bars going up** = the institute's top 10 got this question right more often than ours. Height = how many more students.

3 **Magenta bars going down** = our top 10 got it right more often.

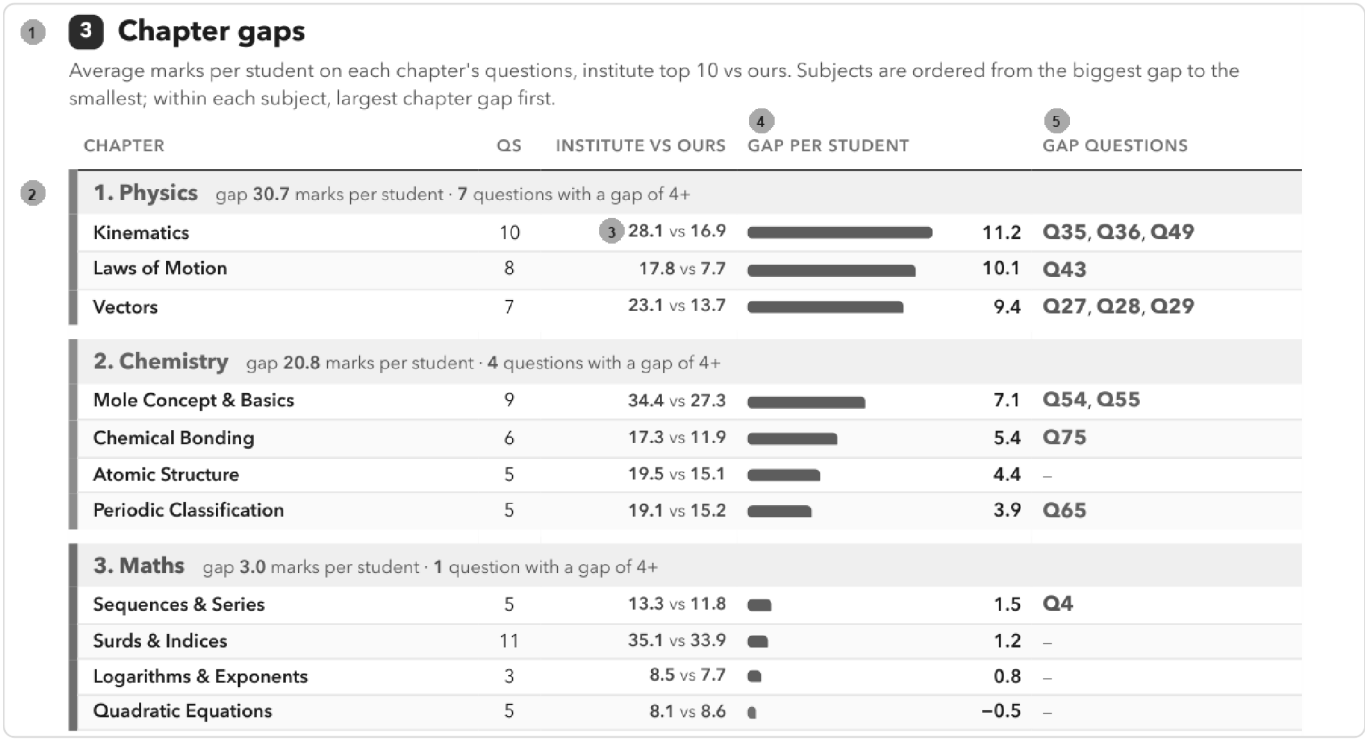
4 **Dashed lines** at +4 and −4 = the gap line. Bars beyond it (and "ahead" bars of 2 or more) are drawn in full colour with their value; the rest are faded.

5 **Panel summary.** Number of gap questions in the subject and the subject's gap per student.

```
gap on a question = (institute number right) − (our number right)    each scaled to "out of 10"
```

CLASS 8 Q35: institute 9 right, ours 1 → +8 (violet, up). Q2: institute 3 right, ours 10 → −7 (magenta, down).

C5 Section 3 – Chapter gaps



1 **3 Chapter gaps**

Average marks per student on each chapter's questions, institute top 10 vs ours. Subjects are ordered from the biggest gap to the smallest; within each subject, largest chapter gap first.

CHAPTER	QS	INSTITUTE VS OURS (3)	GAP PER STUDENT (4)	GAP QUESTIONS (5)
2 **1. Physics** gap **30.7** marks per student · **7** questions with a gap of 4+				
Kinematics	10	28.1 vs 16.9	11.2	Q35, Q36, Q49
Laws of Motion	8	17.8 vs 7.7	10.1	Q43
Vectors	7	23.1 vs 13.7	9.4	Q27, Q28, Q29
2. Chemistry gap **20.8** marks per student · **4** questions with a gap of 4+				
Mole Concept & Basics	9	34.4 vs 27.3	7.1	Q54, Q55
Chemical Bonding	6	17.3 vs 11.9	5.4	Q75
Atomic Structure	5	19.5 vs 15.1	4.4	–
Periodic Classification	5	19.1 vs 15.2	3.9	Q65
3. Maths gap **3.0** marks per student · **1** question with a gap of 4+				
Sequences & Series	5	13.3 vs 11.8	1.5	Q4
Surds & Indices	11	35.1 vs 33.9	1.2	–
Logarithms & Exponents	3	8.5 vs 7.7	0.8	–
Quadratic Equations	5	8.1 vs 8.6	−0.5	–

Section 3 – chapter gaps grouped by subject

1. **Section title.**
2. **Subject block.** Subjects in order of their gap (biggest first), with the subject gap and how many gap questions it has. Chapters within a subject run from biggest gap to smallest.
3. **Institute vs ours** – average marks per student on that chapter's questions.
4. **Gap per student** = the difference, with a bar on one shared scale (violet = we are behind, magenta = we are ahead, shown with a minus sign).
5. **Gap questions** – the chapter's questions that crossed the gap line.

CLASS 8 Physics · Kinematics: 28.1 vs 16.9 → gap 11.2 marks per student; gap questions Q35, Q36, Q49.

C6 Section 4 – Biggest gaps, question list

4 Biggest gaps – question list

Questions where at least 4 more of the institute's top 10 got the right answer than ours, one table per subject. The cropped question for each is in section 5.

Maths

1 question with a gap of 4+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q4	**Sequences & Series** – Identify AP / GP / HP from a surd expression	Single correct · +4 / −1	2 8/10	3 3/10 · 5 · 2	+5 4

Physics

7 questions with a gap of 4+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q35	**Kinematics** – a = kv² – integrate to get v(s)	Single correct · +4 / −1	9/10	1/10 · 1 · 8	+8
Q43	**Laws of Motion** – Normal reaction across an inclined contact	Single correct · +4 / −1	7/10	1/10 · 7 · 2	+6
Q28	**Vectors** – Resultant ⊥ smaller vector; find magnitudes	Single correct · +4 / −1	10/10	5/10 · 4 · 1	+5
Q29	**Vectors** – Angles between three vectors summing to zero	Single correct · +4 / −1	10/10	5/10 · 4 · 1	+5
Q27	**Vectors** – Vector components under axis rotation	Single correct · +4 / −1	7/10	3/10 · 2 · 5	+4
Q36	**Kinematics** – a(t) with modulus – area under a-t	Single correct · +4 / −1	5/10	1/10 · 9 · 0	+4
Q49	**Kinematics** – Free fall then deceleration – find height	Numerical · +4 / −1	5/10	1/10 · 2 · 7	+4

Chemistry

4 questions with a gap of 4+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q55	**Mole Concept & Basics** – Laws of chemical combination	Single correct · +4 / −1	10/10	5/10 · 4 · 1	+5
Q75	**Chemical Bonding** – Hybrid orbitals + lone pairs in NH_3	Numerical · +4 / −1	8/10	3/10 · 4 · 3	+5
Q54	**Mole Concept & Basics** – Minimum molecular mass of insulin from 3.4% S	Single correct · +4 / −1	10/10	6/10 · 1 · 3	+4
Q65	**Periodic Classification** – Largest radius among isoelectronic ions	Single correct · +4 / −1	8/10	4/10 · 5 · 1	+4

5 **Where our top 10 did better**

Q2 10 vs 3 right | Q8 10 vs 7 right | Q24 8 vs 5 right | Q3 5 vs 3 right

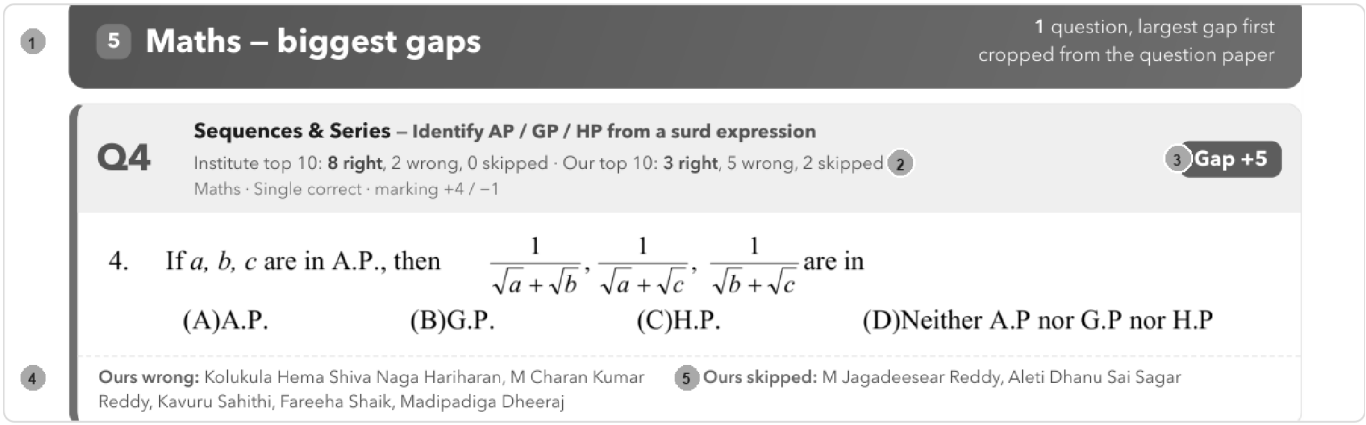
6 **How to read this report.** "Institute top 10" = the *Top 10 Students* rows of the institute's question-wise R/W/U sheet for this test (program ranks 1–10 in Pragya G8 (EXI) Achievers, all campuses). "Our top 10" = this campus's top 10 by program rank. A gap of +5 on a question means 5 more of the institute's top 10 got it right than ours.
Final key applied. Deleted in the final key: Q20. Answer revised in the final key and counted as marked in the final marks sheet: Q49 (key 298), Q67 (key D). Left-out questions are marked "del" or "n/a" in the charts and excluded from both groups.

Section 4 – the gap questions, one table per subject

1. **Section title.**
2. **Institute right** – how many of the institute's top 10 got it right (of 10).
3. **Ours right · wrong · skipped** – our top 10's full split, so you can see whether we lost it by guessing wrong or by skipping.
4. **Gap** pill – how many more of the institute's top 10 got it right.
5. **Where our top 10 did better** – every question where 2 or more extra of ours got it right, with both counts.
6. **Notes** – the definitions of both groups and the final-key treatment.

Other columns are the same as in the impact report: question number, chapter and concept, type and marking. Rows are ordered by gap, biggest first.

C7 Section 5 – Question cards



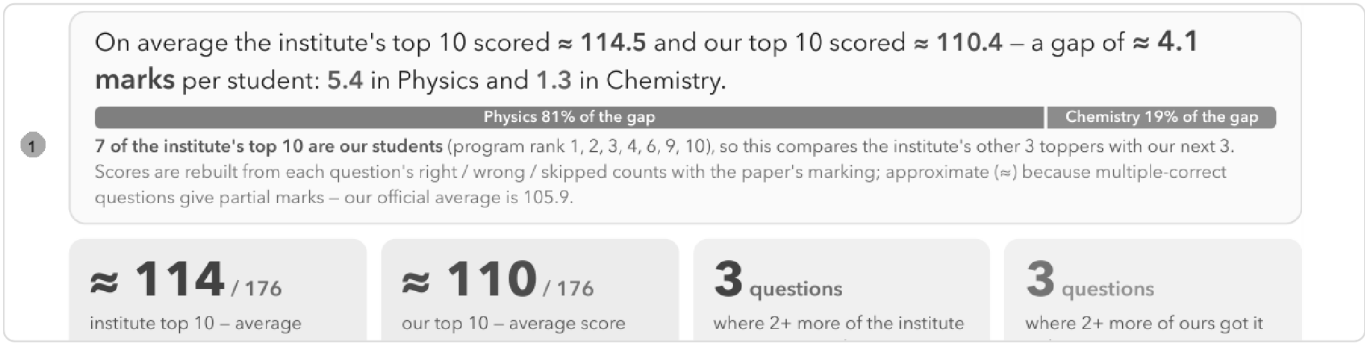
Section 5 – a gap-question card

1. **Subject banner** – each subject on a new page, biggest gap first.
2. **Both groups' counts** – institute top 10 right / wrong / skipped, then ours.
3. **Gap** pill.
4. **Ours wrong** – names of our students who answered it wrong.
5. **Ours skipped** – names of our students who skipped it.

Questions that are both high-impact and gap questions appear in both reports; the crop is identical.

C8 Special cases

When most of the institute's top 10 are our own students

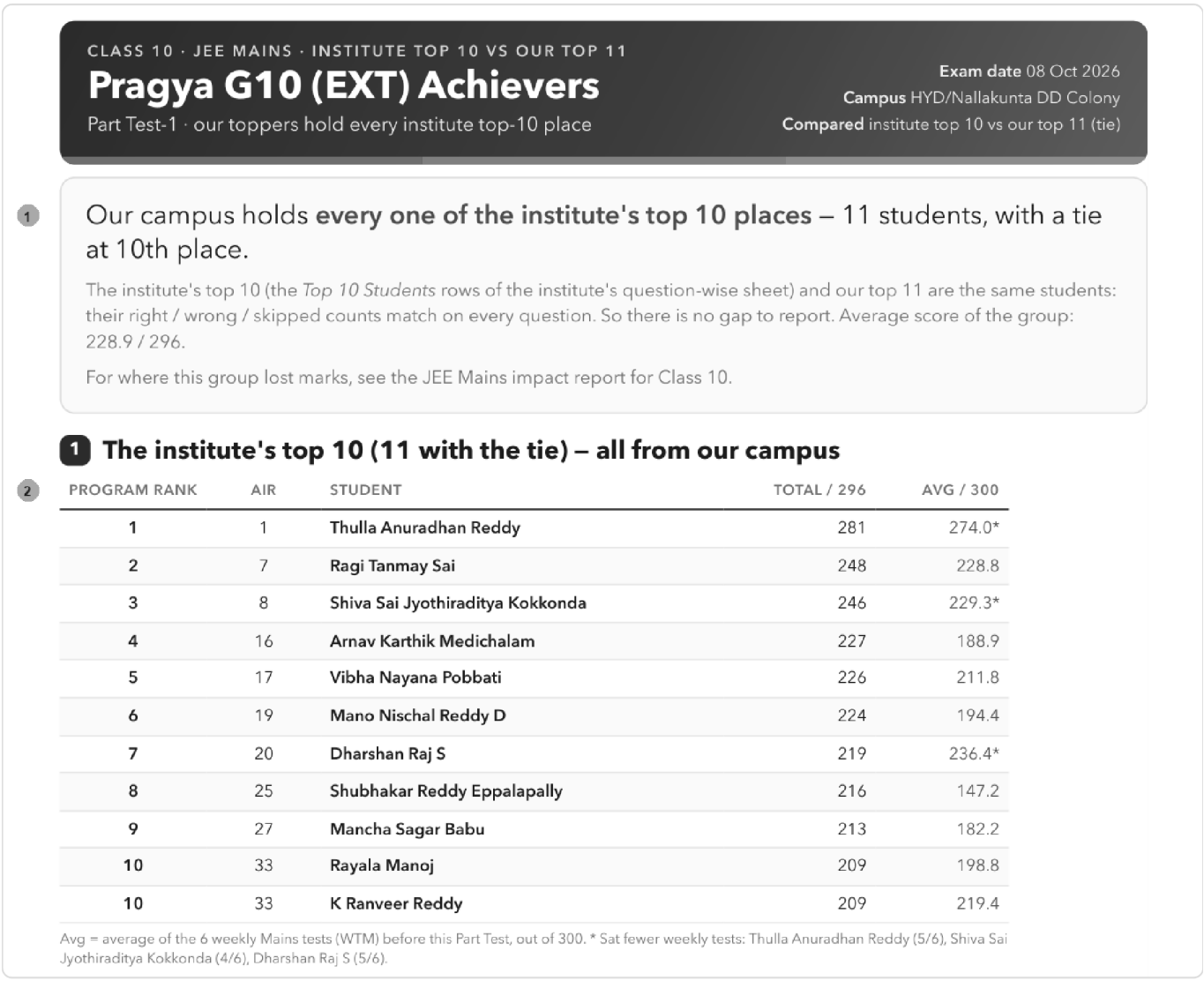


G9 Advanced comparison – the overlap line

1. **Overlap line.** Lists the program ranks of our students inside the institute's top 10. Those students are in both groups, so they cancel out: the comparison is really "the institute's other toppers vs our next ones".

With 7 of 10 shared, at most 3 answers can differ on any question, so a gap of 4 is almost impossible. When at most 5 students differ, the report lowers the gap line from 4 to **2** and says so. This applies to Class 9 (both exams) and Class 10 Advanced; Class 8 uses 4.

When our top 10 *are* the institute's top 10



CLASS 10 · JEE MAINS · INSTITUTE TOP 10 VS OUR TOP 11

Pragya G10 (EXT) Achievers

Part Test-1 · our toppers hold every institute top-10 place

Exam date 08 Oct 2026
Campus HYD/Nallakunta DD Colony
Compared institute top 10 vs our top 11 (tie)

1 Our campus holds **every one of the institute's top 10 places** – 11 students, with a tie at 10th place.

The institute's top 10 (the *Top 10 Students* rows of the institute's question-wise sheet) and our top 11 are the same students: their right / wrong / skipped counts match on every question. So there is no gap to report. Average score of the group: 228.9 / 296.

For where this group lost marks, see the JEE Mains impact report for Class 10.

1 The institute's top 10 (11 with the tie) – all from our campus

2

PROGRAM RANK	AIR	STUDENT	TOTAL / 296	AVG / 300
1	1	Thulla Anuradhan Reddy	281	274.0*
2	7	Ragi Tanmay Sai	248	228.8
3	8	Shiva Sai Jyothiraditya Kokkonda	246	229.3*
4	16	Arnav Karthik Medichalam	227	188.9
5	17	Vibha Nayana Pobbati	226	211.8
6	19	Mano Nischal Reddy D	224	194.4
7	20	Dharshan Raj S	219	236.4*
8	25	Shubhakar Reddy Eppalapally	216	147.2
9	27	Mancha Sagar Babu	213	182.2
10	33	Rayala Manoj	209	198.8
10	33	K Ranveer Reddy	209	219.4

Avg = average of the 6 weekly Mains tests (WTM) before this Part Test, out of 300. * Sat fewer weekly tests: Thulla Anuradhan Reddy (5/6), Shiva Sai Jyothiraditya Kokkonda (4/6), Dharshan Raj S (5/6).

G10 Mains comparison – the one-page version

1 **Headline.** If both groups have identical right / wrong / skipped counts on every question, there is no gap to report and the comparison becomes a single page.

2 **Student list** – program rank, AIR, name, total and weekly average of every student in the institute's top 10 (11 with the tie).

PART D

Formulas and thresholds

Every calculation in the reports, in one place. “n” = number of students in the group (10, or 11 with a tie). “marks” and “negative” are the question's own marking from A4.

D1 Every formula

Per question (impact report)

```
lost by            = number wrong + number skipped                          (0 … n)
high-impact        = lost by ≥ 5
level              = Critical if lost by ≥ n−1 · High if ≥ n−3 · otherwise Moderate
marks lost         = wrong × (marks + negative) + skipped × marks
negative marks     = wrong × negative
```

Per subject (impact report)

```
marks lost         = Σ marks lost of the subject's valid questions
  on wrong         = Σ wrong × (marks + negative)       on skipped = Σ skipped × marks
per student        = marks lost ÷ n
share              = subject marks lost ÷ all marks lost
accuracy           = right ÷ (right + wrong)                                (skips ignored)
type % wrong       = wrong answers in that section ÷ (section questions × n)
subject maximum    = subject marks on the paper − marks of left-out questions
```

Per chapter (impact report)

```
marks lost         = Σ marks lost of the chapter's questions
per student        = chapter marks lost ÷ n
students per Q     = Σ lost by ÷ number of questions in the chapter
order              = subjects by marks lost (most first), then chapters by marks lost
```

Per student (impact report)

```
subject score      = subject maximum − Σ wrong × (marks + negative) − Σ skipped × marks
percentage         = subject score ÷ subject maximum
weakest subject    = subject with the lowest percentage
lost (grid)        = number of high-impact questions this student got wrong or skipped
Avg                = taken from the averages sheet  ·  ∗ = sat fewer tests than held
```

Headline (impact report)

```
total marks lost   = Σ subject marks lost
tally (Mains)      = n × valid paper maximum − Σ official totals   (= total marks lost)
top-10 average     = Σ official totals ÷ n
```

Comparison report

```
group marks on Q   = marks × right − negative × wrong
group average      = Σ group marks ÷ group size                (per subject and overall)
gap per student    = institute average − our average
gap share          = subject gap ÷ Σ positive subject gaps
gap on Q           = institute right × 10 ÷ group size − our right × 10 ÷ n
gap question       = gap on Q ≥ gap line     (gap line = 4, or 2 when ≤ 5 students differ)
ahead question     = gap on Q ≤ −2
right/wrong/skip % = count ÷ (subject questions × group size)
chapter gap        = institute chapter average − our chapter average
```

CLASS 8 Every formula above is checked automatically on every build: e.g. Mains subject scores must add up to each official total, and the institute's Top 10 rows must cover exactly 10 students (11 with a tie).

D2 Every threshold, and where it lives

These are the only "judgement" numbers in the tool. Changing one changes every report on the next build.

THRESHOLD	VALUE	IN THE TOOL	WHAT IT CONTROLS
High-impact line	5 of the top 10	`analyze.py · HIGH_IMPACT_MIN`	A question lost (wrong or skipped) by this many or more of the top 10 is "high-impact".
Critical / High / Moderate	lost by n−1…n / n−3…n−2 / 5…n−4	`context.py · tier()`	Colour of the "Lost by" pill. With n = 10: 9–10, 7–8, 5–6.
Question-type call-out	1.5 × the paper's overall loss rate	`context.py · _worst_question_type()`	A takeaway names a question type only if it is lost at least 1.5 times as often as the paper overall.
Skipping call-out	15% of answers blank	`context.py · takeaways()`	A takeaway names the most-skipped section only if 15% or more of its answers were left blank.
Comparison gap line	4 more students right	`compare.py · GAP_MIN`	A question is a "gap question" when this many more of the institute top 10 got it right than ours.
Gap line when groups overlap	2 more students right	`compare.py · GAP_MIN_OVERLAP`	Used instead when at most 5 of the institute top 10 are not our students.
"Where we did better"	2 more of ours right	`compare.py · AHEAD_MIN`	Questions our top 10 got right at least this many more times than the institute top 10.
Comparison skip / wrong call-out	5 percentage points	`compare.py · _takeaways()`	A takeaway mentions skipping or wrong answers only when ours is 5+ points worse than the institute's.
"We are ahead in …"	0.5 marks per student	`compare.py · _takeaways()`	A subject is called out as a lead only when our average beats the institute's by more than 0.5.
Short names	names longer than 24 letters	`context.py · SHORT_NAME_MAX`	Long names get middle-name initials in the tight tables (full names stay on the cards).
Top students per campus	10 (+ ties)	`parse_marks.py · TOP_N`	Everyone tied with the 10th program rank is included, so a group can be 11.
Question crop resolution	200 dpi	`crop.py · DPI`	Questions are cut from the paper at this sharpness.

PART E

Glossary A–Z

Every word, label and symbol you will meet in the reports. Words in *italics* have their own entry.

#

≈

Approximate. Used on Advanced marks figures because multiple-correct questions give partial marks that the marks sheet does not show. Counts of students are never approximate.

*

After an average (Avg): the student sat fewer weekly tests than were held. The notes list each such student as "Name (sat/held)".

✓ ✗ –

Right, wrong, skipped — in the student grid (section 5) and wherever answers are shown per student.

A

Accuracy

Right ÷ (right + wrong): how often a student's attempted answers were correct. Skips do not count.

Ahead question

Comparison report: a question that at least 2 more of our top 10 got right than the institute's top 10. Listed in "Where our top 10 did better".

AIR

All-India Rank — a student's rank across the whole class programme (all batches, all campuses), from the All India Marks Analysis sheet. Not the same as program rank.

Answered wrong by 5+

Box on page 1: questions with 5 or more wrong answers. Signals a misconception or trap.

Avg

The student's average over the weekly tests before this Part Test: WTM out of 300 (Mains), or WTA scaled from 240 to 180 (Advanced) so it matches the paper. From the averages workbook.

B

Batch

The program group the students belong to, e.g. Pragya G8 (EXI) Achievers.

C

Cards

Section 7 of the impact report / section 5 of the comparison: one block per question with the cropped question and the names of who lost it.

Chapter

The syllabus chapter a question belongs to, assigned by reading the question.

Concept note

The short "what it tests" line after a chapter name, e.g. "Strongest ionic bond (charge and size)".

Critical / High / Moderate

Severity of a high-impact question: lost by 9–10, 7–8, or 5–6 of the top 10 (with 11 students: 10–11, 8–9, 5–7).

D

del

Deleted: the question is DEL in the final key (or has no entry for anyone when there is no final key). Left out of everything; its number is struck through in charts.

F

Final key

The corrected answer key issued after the test. Its DEL questions are removed from every count.

Final marks sheet

The institute's "Toppers Vs Individual Student Qwise RWU Analysis" Excel file — the main data source.

G

Gap (comparison)

How many more of the institute's top 10 got a question right than ours (per question), or how many more marks per student they scored (per subject / chapter).

Gap line

The gap at which a question counts as a gap question: 4, or 2 when most of the institute's top 10 are our students.

Gap question

A question at or above the gap line.

H

High-impact

A question lost by 5 or more of our top 10.

High-impact line

The dashed line at 5 in the impact charts.

I

Institute top 10

The 10 best students in the program across all campuses (program ranks 1–10).

L

Left blank by 5+

Box on page 1: questions skipped by 5 or more. Signals an unfamiliar topic, low confidence or time pressure.

Lost

Not got right — answered wrong or skipped.

Lost by

> How many of the top 10 lost a question.

M

Marks lost

> Marks the group gave away: for each wrong answer the question's marks plus its negative mark, for each skip the question's marks.

N

n/a

> Not available: a question that is valid in the final key but has no right / wrong / skipped entry for anyone, so it cannot be counted.

Negative marks

> Marks deducted for wrong answers (−1 in Mains; −1 or −2 in some Advanced sections; none in Integer and Paragraph).

Numerical

> Mains question type Q21–25 (and the same positions in Physics and Chemistry): a number answer, +4 / −1.

O

Our top 10

> This campus's 10 best students by program rank (11 with a tie).

P

Paragraph

> Advanced question type built on a shared passage (two questions per passage), +3 / 0. The passage is shown on the card.

Per student

> A group total divided by the number of students.

Program rank

> Rank within the batch across all campuses, from the final marks sheet (where the column is labelled AIR or ARI). Used to pick our top 10.

R

R / W / U

> Right / Wrong / Unanswered — the three marks in the final marks sheet.

Rank & score

> The four columns beside each name in section 5: Prog. rank, AIR, Total, Avg.

S

Share

> A subject's marks lost as a percentage of all marks lost.

Single correct

> Question type with one right option: +4 / −1 in Mains, +3 / −1 in Advanced.

Skipped

> Left blank (U).

Students per Q

> On average, how many of the top 10 lost each question in a chapter.

Students who lost it

> The bottom row of the student grid: per question, how many lost it.

Subject maximum

> The subject's marks on the paper minus left-out questions, e.g. Maths 96 = 100 − Q20 (4).

T

Takeaways

> Automatically written sentences under section 1; each has a rule (B3, C3).

Tie

> Students with the same program rank as the 10th student are all included, so a top 10 can be 11.

Total

> The student's official marks in this test, as printed in the final marks sheet.

W

Weakest

> The subject with the lowest percentage of its maximum for that student.

WTM / WTA

> Weekly Test Mains / Weekly Test Advanced — the tests the averages are taken from.

PART F

Behind the scenes: the tool

The reports are produced by a small set of Python scripts in `_report_tools`. Nothing is edited by hand, so the same inputs always give the same reports. This part is for whoever maintains them.

F1 Folder map

```
SCHOOL PART TEST REPORTS/
├── FINAL MARKS/                 final marks sheets (MAINS MARKS, ADV MARKS) — main data source
├── MAINS REPORTS/, ADV REPORTS/ campus result sheets (PDF) — batch name, cross-check
├── QUESTION PAPERS/             MAINS, ADVANCE — question papers (PDF) for the crops
├── PART TEST-1 ANALYSIS-COMMON/ All India Marks Analysis — AIR
├── HYD_…_AVERAGES (1).xlsx      weekly-test averages — Avg
├── MOST IMPACT QUESTIONS/       G8, G9, G10 — reports exactly as built
├── FINAL REPORTS/               G8, G9, G10 — reports to share, plus this book
└── _report_tools/               the tool (below); build/ holds working files
```

F2 What each file does

FILE	JOB
`paper_config.py`	The settings for each class and exam: file locations, paper pattern, final key, chapter tags, concept notes, crop fixes, averages sheet (F3).
`schemes.py`	The paper patterns: which questions belong to which subject and section, and their marking.
`parse_marks.py`	Reads a final marks sheet: every student's R/W/U per question, program rank, total; picks our top 10 (+ ties); reads the institute Top 10/20/50/100 rows. Accepts the rank column labelled AIR or ARI.
`parse_report.py`	Reads a campus result-sheet PDF: batch name, exam date, campus.
`parse_air.py`	Reads the All India Marks Analysis sheets and looks each student up by admission number.
`parse_avg.py`	Reads the averages workbook (WTM and WTA blocks, tests sat and held).
`analyze.py`	Applies the final key, then counts lost / wrong / skipped and marks lost per question, subject, section and student.
`context.py`	Turns the counts into everything the impact report shows: tables, tiles, chapters, cards and the takeaway sentences.
`compare.py`	The same for the comparison report: group scores, gaps, gap line, overlap handling and its sentences.
`charts.py`	Draws the bar charts (section 2 of both reports).
`crop.py`	Cuts each question out of the question paper (F4).
`template.html`, `compare.html`, `base_styles.html`	The page layouts and styles of the two reports.
`build_report.py`	Runs everything and prints the PDFs into `MOST IMPACT QUESTIONS`.
`test_reports.py`	The automatic checks (F5).
`guide/`	Builds this book from the same data (`build_guide.py`, `shot_list.py` for the screenshots).

F3 Paper settings (one block per class and exam)

SETTING	WHAT IT IS	CLASS 8 MAINS VALUE
`key`	Name of the report	G8_Mains
`class_label`, `exam_label`, `test_label`	Wording on the report	Class 8 · JEE Mains · Part Test-1
`marks_xlsx`	The final marks sheet	`FINAL MARKS/MAINS MARKS/…G8 (EXI)_Ach…xlsx`
`report_pdf`	The campus result sheet	`MAINS REPORTS/G8_…pdf`
`paper_pdf`, `last_q`	The question paper and its last question number	Q1–Q75
`pattern`	Mains or Advanced pattern (A4)	Mains
`marks_exact`	True for Mains; False for Advanced (adds ≈)	True
`final_key`	DEL questions, revised answers, and valid questions with no data	DEL Q20
`chapters`	Chapter of every question	75 questions tagged
`concepts`	"What it tests" note for questions shown in tables and cards	34 notes
`extra_stems`, `cuts`	Crop fixes for unusual layouts (F4)	none needed
`avg_sheet`	Sheet of the averages workbook	G8_ACH

F4 How question images are cut

1. Find every question number ("1.", "2.", …) in the paper's left margin, in reading order, and every section heading, instruction block and paragraph heading.
2. A question runs from its number down to the next number or heading — across pages if needed.
3. The cut is moved to the nearest blank strip so formulas, fractions and diagrams are not sliced. If a gap is too tight it cuts just above the number instead.
4. For a multiple-choice question, anything after its last option "D)" and before the next number belongs to the *next* question (some papers print a figure or reaction above the question number).
5. Paragraph / passage questions get their passage placed above them.
6. The page border and full-width rules are removed; the colour is kept.

Manual fixes in use (each found by checking every crop by eye):

PAPER	FIX	WHY
Class 8 Advanced	Passage starting "Sum of n terms…" marked as the paragraph for Q14–15	It has no "PARAGRAPH" heading
Class 9 Advanced	Cut above Q9 at 435.5 pt	The Σ upper limit of Q9 sits level with Q8's last line
Class 10 Mains	Cut above Q75 at 210 pt	Q75's reaction scheme is printed above its number

F5 Automatic checks (run on every paper)

CHECK	WHAT IT PROTECTS AGAINST
Reads 10 students (11 with a recorded tie) with all three subjects	A wrong or changed spreadsheet layout
Right + wrong + skipped = number of valid questions, per student and subject	Questions dropped or double-counted
Mains: rebuilt subject scores add up to every official total	Any error in the marking scheme or marks-lost formula
Every question has a chapter	A forgotten chapter tag
Every high-impact question is lost by 5+	Threshold mistakes
Every question number is found in the paper	Crops missing or shifted
Final marks sheet top 10 = result sheet top 10 (ranks and totals)	Using the wrong batch's sheet
Final-key DEL questions are excluded and never high-impact	The final key not being applied
Institute Top 10 rows cover exactly 10 students (or the tie size)	Reading the wrong summary row
Every top student has an AIR and (where set) an average	Missing or mismatched admission numbers

F6 Rebuilding, and the next Part Test

To rebuild everything (e.g. after changing a chapter tag), in Terminal:

```
cd ~/"Desktop/SCHOOL PART TEST REPORTS/_report_tools"
python3 -m pytest -q test_reports.py        # checks — should all pass
python3 build_report.py                     # all 12 reports → MOST IMPACT QUESTIONS
python3 build_report.py G8_Mains            # just one report
python3 guide/build_guide.py                # this book → FINAL REPORTS
```

> **Copying reports to share.** PDFs written by the browser carry a macOS "quarantine" tag that can stop them opening on some Macs. Copy them with `ditto --noextattr --noqtn <source> <destination>` (this book is copied that way automatically).

For the next Part Test:

1. Put the new question papers, final marks sheets, All India sheets and averages workbook in the same folders.
2. In `paper_config.py`, update each block's file paths, the final key, and the chapter tags and concept notes for the new questions.
3. Run the checks; fix anything they report (most often: a chapter tag missing, or a question number not found in the paper).
4. Build, then look through every question card — a new paper layout may need a crop fix (F4).
5. Copy the reports to `FINAL REPORTS` with `ditto`.

PART G

Data notes and FAQ

What the reports cannot know, where the source data disagrees with itself, and answers to questions that came up while the reports were being made.

G1 Known limitations and data notes

1. **Advanced marks are approximate (≈).** Partial credit on multiple-correct questions is not in the marks sheet. All student counts are exact.

2. **The final marks sheet may not be fully regraded with the final key.** In the Class 8 Advanced sheet, nobody is marked right on Q48 (final key 2 OR 5) — 0 right among the institute's top 100 — or on Q51 (key 3), 0 right among the top 100. In Class 8 Mains, Q67 (key D) is marked wrong for all of our top 10. The reports count these as marked (rule 2 in A7), so they may look worse than they really are. A sheet regraded with the final key would fix this automatically.

3. **Q13 and Q29 (Class 8 Advanced) cannot be counted.** They are valid in the final key, but no student — in our campus or institute-wide — has a right / wrong / skipped entry for them. They are shown as "n/a" and taken off the subject maxima.

4. **All-India sheet totals differ for some Advanced students.** The All India Marks Analysis gives different totals from the final marks sheet for 10 of our Class 8 Advanced top 10 (e.g. Ruperao Shritan 117 → 125, Kolukula Hema Shiva Naga Hariharan 116 → 121) and 2 of Class 9 Advanced. It looks regraded. The reports use the final marks sheet totals, because those match the question-by-question data; the AIR column comes from the All India sheet.

5. **No final keys for Classes 9 and 10.** Only questions with no entry for anyone are left out there (A7).

6. **Chapter tags and concept notes are judgement.** They were assigned by reading each question and may differ from your department's classification.

7. **Weekly averages cover different numbers of tests** for students who missed tests (marked *). The averages workbook lists test 6 of the Mains series as "19-09-2025" — almost certainly 2026; it does not affect the numbers.

8. **Paper cover vs paper body.** The Class 8 Advanced cover puts Chemistry Section I at Q35–37, the paper prints Q35–38. The paper is followed.

G2 Questions people have asked

Q. What are the numbers in the chapter table (section 3)?

Marks lost: the total marks all 10 students together gave away on that chapter's questions. In Mains a wrong answer costs 5 (4 not earned + 1 negative) and a skip costs 4. "Per student" divides by 10; "Students per Q" is how many of the 10 lost each question on average. See B5.

Q. Why does a chapter say "None reached 5"?

None of its questions was lost by 5 or more of the top 10, so it has no high-impact question. The report names its most-missed question(s) instead — e.g. Atomic Structure in Class 8 Mains: Q57 and Q58, each lost by 3.

Q. Why is Q13 marked deleted / "n/a" in Class 8 Advanced?

The marks sheet has no entry for anyone on Q13 (and Q29). The final key gives answers (4046 and 9), so they are not "deleted" but "n/a" — they cannot be counted. Q14, Q17 and Q50 are deleted (DEL) in the final key. See A7 and G1.

Q. Why is Advanced Maths "out of 50", Physics 56, Chemistry 57?

Each subject is worth 60, minus the questions left out: Maths 60 − Q13 (4) − Q14 (3) − Q17 (3) = 50; Physics 60 − Q29 (4) = 56; Chemistry 60 − Q50 (3) = 57. The reports print this subtraction under each subject.

Q. Why are Advanced figures marked ≈?

Partial marks on multiple-correct questions are not in the data. Counts of students are exact; marks are close but not exact.

Q. Is "Prog. rank" the same as AIR?

No. Program rank is within the batch (e.g. Pragya G8 (EXI) Achievers) across all campuses; the final marks sheet labels it "AIR", but it is really this program rank. AIR is the real All-India rank across the whole class programme. Example: Kolukula Hema Shiva Naga Hariharan is program rank 11 but AIR 26.

Q. Why does an average have a * next to it?

The student sat fewer weekly tests than were held, so the average is over fewer tests. The notes list them as "Name (sat/held)".

Q. Why is the Advanced Avg out of 180 when the weekly tests are out of 240?

The weekly Advanced tests (WTA) are marked out of 240 (80 per subject). The reports scale every average by 180/240 so it is on the same 180-mark scale as this paper (60 per subject). Example: an average of 176.9 / 240 becomes 132.7 / 180 – both 73.7%.

Q. Why does Class 10 Mains have 11 students?

Two students tie at program rank 10, and both are included. All "of 10" wording becomes "of 11".

Q. Why do the Class 9 (and Class 10 Advanced) comparisons use a gap of 2 instead of 4?

Seven of the institute's top 10 are our own students, so at most 3 answers can differ on any question and a gap of 4 is almost impossible. The gap line drops to 2 whenever 5 or fewer of the institute's top 10 are not ours. See C8.

Q. Why is the Class 10 Mains comparison only one page?

Our top 11 *are* the institute's top 10 (11 with the tie): every count matches, so there is no gap to show. The page lists the students instead.

Q. What is the difference between "Lost by" and "Lost" in the student grid?

"Lost by" (per question) = how many students lost that question. "Lost" (per student, last grid column) = how many of the high-impact questions that student lost.

Q. Can I trust the cropped questions?

Every crop in all 12 reports was checked by eye; three unusual layouts needed manual fixes (F4). If a crop ever looks wrong, the paper layout can be fixed in the settings and the reports rebuilt.

Q. A file or zip will not open on my Mac.

A zip downloaded through chat can arrive cut off (the original was 16.4 MB, the broken download 0.9 MB) – use the copies in `FINAL REPORTS` instead. PDFs printed by the browser carry a macOS "quarantine" tag; the copies in `FINAL REPORTS` have it removed. If one still refuses, right-click → Open With → Preview.

Q. How do I change a chapter tag or add a final key?

Edit that paper's block in `_report_tools/paper_config.py` (the `chapters` or `final_key` setting), then rebuild (F6). Every number, sentence and chart updates automatically.